



Final
MISO DPP 2019 Cycle 1 South Area
Study Phase I Report

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MISO
720 City Center
Drive Carmel
Indiana 46032
<http://www.misoenergy.org>

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1. Executive Summary

This report presents the results of a System Impact Study performed to evaluate the interconnection of the generators in the DPP 2019 Cycle 1 South Area Phase I (South Area DPP 1). The study was performed under the direction of MISO and reviewed by an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Entergy, Cooperative Energy, CLECO, AECC, AECl, Southern Company, PowerSouth, SPP, and TVA.

1.1. Project List

The original interconnection requests for DPP 2019 Cycle 1 South Area had a total of 70 generation projects. Projects J1283, J1291, J1309, J1433, J1443, and J1484 withdrew prior to the start of DPP Phase I. Therefore, there are 64 generation projects with a combined nameplate rating of 9190.18 MW of Energy Resource Interconnection Service (ERIS) & 9059.2 MW of Network Resource Interconnection Service (NRIS). The detailed list of South Area DPP 1 is shown below in Table 1, and the Phase I study was kicked off on February 10th, 2020.

Table 1: List of DPP 2019 Cycle 1 South Area Phase I Projects

Project	Fuel Type	Transmission Owner	County	State	Service Requested	MW	Point of Interconnection (POI)
J1205	Solar	Cleco Corporation	Allen Parish, Assumption Parish	LA	NRIS	125	Centennial - West Bay 138 kV
J1215	Solar	Entergy Arkansas	Arkansas County	AR	NRIS	200	Stuttgart Ricuskey – Woodward 230 kV
J1217	Solar	Cleco Corporation	St. Landry Parish	LA	NRIS	200	Ville Platte – West Fork 230 kV
J1219	Solar	Entergy Louisiana	Tangipahoa Parish	LA	NRIS	100	Chicken Farm – Amite 115 kV
J1220	Solar	Cleco Corporation	Rapides Parish	LA	NRIS	200	Rapides – Rodemacher 230 kV
J1223	Solar	Entergy Mississippi	Pike County	MS	NRIS	150	Colonial Pipeline – Kentwood 115 kV
J1236	Battery Storage	Cleco Corporation	St. Landry Parish	LA	NRIS	50	Ville Platte – West Fork 230 kV
J1238	Battery Storage	Cleco Corporation	Rapides Parish	LA	NRIS	50	Rapides – Rodemacher 230 kV
J1239	Solar	Entergy Louisiana	Ouachita Parish	LA	NRIS	100	Rilla 115 kV Substation
J1245	Battery Storage	Entergy Arkansas	Arkansas County	AR	NRIS	50	Stuttgart Ricuskey – Woodward 230 kV
J1246	Gas	Entergy Louisiana	Iberville Parish	LA	NRIS	725.2	Bayou Labutte 500 kV Substation
J1247	Waste Heat Recovery	Cleco Corporation	St Mary Parish	LA	ERIS	30.98	Ivanhoe 138 kV Substation
J1249	Gas	Cooperative Energy	Lamar County	MS	NRIS	600	Morrow 161 kV Substation
J1250	Battery Storage	Entergy Arkansas	Clay County	AR	NRIS	35	Datto – Corning North 161 kV
J1257	Battery Storage	Entergy Louisiana	Tangipahoa County	LA	NRIS	20	Chicken Farm – Amite 115 kV
J1258	Battery Storage	Entergy Arkansas	Lonoke County	AR	NRIS	50	KEO EHV 500 kV Substation
J1259	Battery	Entergy	Pike County	MS	NRIS	50	Colonial Pipeline – Kentwood 115 kV

Project	Fuel Type	Transmission Owner	County	State	Service Requested	MW	Point of Interconnection (POI)
	Storage	Mississippi					
J1260	Solar	Entergy Arkansas	Desha County	AR	NRIS	164	Dumas – Reed 115 kV
J1261	Solar	Entergy Arkansas	Monroe County	AR	NRIS	90	Clarendon – Brinkley East 115 kV
J1267	Solar	Entergy Arkansas	Cross County	AR	NRIS	135	Fisher – Cherry Valley 161 kV
J1277	Solar	Entergy Texas	Liberty County	TX	NRIS	130	Pintail 138 kV Substation
J1278	Solar	Entergy Texas	Polk County	TX	NRIS	100	Corrigan Bulk – Staley 138 kV
J1279	Solar	Entergy Texas	Hardin County	TX	NRIS	100	Rye – Cypress 138 kV
J1280	Solar	Entergy Arkansas	Monroe County	AR	NRIS	100	Clarendon – Brinkley East 115 kV
J1281	Solar	Entergy Arkansas	Ashley County	AR	NRIS	75	Crossett – Monticello East 115 kV
J1282	Solar	Cleco Corporation	Beauregard Parish	LA	NRIS	200	Cooper – Penton Road 230 kV
J1286	Solar	Cleco Corporation	Evangeline Parish	LA	NRIS	150	Manuel 138 kV Substation
J1287	Solar	Entergy Mississippi	Tunica County	MS	NRIS	50	Tunica – Crenshaw 115 kV
J1290	Battery Storage	Cleco Corporation	Evangeline Parish	LA	NRIS	25	Manuel 138 kV Substation
J1293	Battery Storage	Entergy Arkansas	Lee County	AR	NRIS	25	Brinkley East – Moro 115 kV
J1294	Solar	Entergy Mississippi	Sunflower County	MS	NRIS	99	Indianola – Cleveland 115 kV
J1347	Solar	Entergy Mississippi	Scott County	MS	NRIS	175	Pelahatchie – Scott 115 kV
J1351	Solar	Entergy Louisiana	Madison Parish	LA	NRIS	120	Tallulah – Delhi 115 kV
J1361	Solar	Entergy Louisiana	Washington Parish	LA	NRIS	50	Lagen Holton – Franklinton 115 kV
J1364	Solar	Entergy Arkansas	Crittenden County	AR	NRIS	150	Marked Tree – Marion 161 kV
J1367	Solar	Cleco Corporation	St. Mary Parish	LA	NRIS	95	Caneland 230 kV Substation
J1368	Solar	Entergy Louisiana	Point Coupee Parish	LA	NRIS	200	Fancy Point – Waterloo 230 kV
J1372	Solar	Entergy Louisiana	Point Coupee Parish	LA	NRIS	100	Fancy Point – Waterloo 230 kV
J1373	Solar	Entergy Arkansas	White County	AR	NRIS	95	Griffithville – Searcy Price 115 kV
J1380	Solar	Cleco Corporation	Beauregard Parish	LA	NRIS	200	Cooper – Penton 230 kV
J1384	Solar	Cleco Corporation	Evangeline Parish	LA	NRIS	200	Ville Platte – West Fork 230 kV
J1400	Wind	Entergy Arkansas	Crittenden County	AR	NRIS	150	Marked Tree – Marion 161 kV
J1402	Battery Storage	Entergy Arkansas	Jackson County	AR	NA	0	Cash – Newport AB 161 kV
J1404	Wind	Entergy Mississippi	Sunflower County	MS	NRIS	167	Drew 115 kV Substation

Project	Fuel Type	Transmission Owner	County	State	Service Requested	MW	Point of Interconnection (POI)
J1414	Solar	Entergy Arkansas	Clay County	AR	NRIS	150	Datto – Corning North 161 kV
J1415	Solar	Entergy Arkansas	Lonoke County	AR	NRIS	250	KEO EHV 500 kV Substation
J1421	Solar	Entergy Louisiana	East Feliciana Parish	LA	NRIS	300	Moler 230 kV Substation
J1424	Solar	Cleco Corporation	Beauregard Parish	LA	NRIS	200	Cooper – Penton 230 kV
J1425	Gas	Entergy Arkansas	Jefferson County	AR	NRIS	114	Pine Bluff International Paper 115 kV Substation
J1431	Solar	Entergy Louisiana	St. James Parish	LA	NRIS	200	Vacherie 230 kV Substation
J1434	Solar	Entergy Arkansas	Monroe County	AR	NRIS	100	Brinkley East – Aubrey 230 kV
J1436	Solar	Entergy Louisiana	Morehouse Parish	LA	NRIS	50	Oak Ridge 115 kV Substation
J1437	Wind	Arkansas Electric Cooperative Corporation	Carroll County	AR	NRIS	180	AECC Dry Creek 161 kV Switchyard Station
J1439	Battery Storage	Entergy Mississippi	Tunica County	MS	NRIS	40	Lula – Tunica 115 kV
J1440	Solar	Entergy Mississippi	Tunica County	MS	NRIS	160	Lula – Tunica 115 kV
J1441	Solar	Entergy Louisiana	Terrebonne Parish	LA	NRIS	200	Schriever 230 kV Substation
J1442	Solar	Entergy Texas	Polk County	TX	NRIS	165	Rich (SHECO) – Livingston 138 kV
J1452	Solar	Cleco Corporation	Red River Parish	LA	NRIS	200	Carroll – Mansfield 138 kV
J1455	Solar	Cleco Corporation	St. Mary Parish	LA	NRIS	150	Teche – Morbihan 138 kV
J1458	Solar	Entergy Mississippi	Washington County	MS	NRIS	200	Ray Braswell – Tinnin – Gerald Andrus 230 kV
J1462	Solar	Entergy Louisiana	Acadia Parish	LA	NRIS	150	Acadia – Scanlan 138 kV
J1463	Solar	Entergy Louisiana	West Baton Rouge Parish	LA	NRIS	150	Addis – Big Cajun 230 kV
J1465	Solar	Entergy Louisiana	Pointe Coupee Parish	LA	NRIS	150	Livonia 138 kV Substation
J1469	Solar	Entergy Arkansas	Crittenden County	AR	NRIS	200	Marion – Marked Tree 161 kV

1.2. Total Network Upgrades

The cost allocation of Network Upgrades for the projects in the DPP 2019 Cycle 1 South Phase I reflects responsibilities for mitigating system impacts. The total cost of Network Upgrades is listed in Table 2 below. The costs for Network Upgrades are planning-level estimates and subject to revision in the facility studies. Details pertaining to the cost allocation has been detailed in Appendix A – Cost Allocation Summary (CEII).

Table 2: Total Cost of Network Upgrades for DPP 2019 Cycle 1 South Phase I Projects

Project	ERIS Network Upgrades (\$)				NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrades (\$)	Total Network Upgrade Cost for Milestone Calculation (\$)	M2 (\$)	M3 (\$)
	Thermal	Stability	Short Circuit	Affected System	Deliverability	TO Network Upgrades	TO – Owned Direct Assigned				
a	b	c	d	e	f	g	h	i	j = b+c+d+f+g+i	\$4,000/MW	10% of (j) from Phase I-M2
J1205	3,413,010	TBD	TBD	TBD	26,220,181	10,120,000		0	39,753,191	500,000	3,475,319
J1215	15,061,575	TBD	TBD	TBD	24,999,142	8,500,000		1,968,059	50,528,776	800,000	4,252,878
J1217	10,709,602	TBD	TBD	TBD	57,644,618	6,350,000		0	74,704,220	800,000	6,670,422
J1219	0	TBD	TBD	TBD	9,772	7,500,000		0	7,509,772	400,000	350,977
J1220	6,074,744	TBD	TBD	TBD	109,640,315	6,350,000		0	122,065,059	800,000	11,406,506
J1223	0	TBD	TBD	TBD	14,437	7,500,000		0	7,514,437	600,000	151,444
J1236	2,677,401	TBD	TBD	TBD	14,411,155	6,350,000		0	23,438,555	200,000	2,143,856
J1238	1,518,686	TBD	TBD	TBD	27,410,079	6,350,000		0	35,278,765	200,000	3,327,876
J1239	0	TBD	TBD	TBD	40,163,247	4,000,000		0	44,163,247	400,000	4,016,325
J1245	800,000	TBD	TBD	TBD	6,244,803	8,500,000		0	15,544,803	200,000	1,354,480
J1246	44,500,000	TBD	TBD	TBD	10,292,953	8,000,000		0	62,792,953	2,900,800	3,378,495
J1247	0	TBD	TBD	TBD	0	0	2,928,018	0	0	124,000	0
J1249	5,847,500	TBD	TBD	TBD	0	1,160,000		447,247	7,454,747	2,400,000	0
J1250	0	TBD	TBD	TBD	555,696	7,500,000		0	8,055,696	140,000	665,570
J1257	0	TBD	TBD	TBD	1,954	7,500,000		0	7,501,954	80,000	670,195
J1258	0	TBD	TBD	TBD	820,271	4,000,000		0	4,820,271	200,000	282,027
J1259	0	TBD	TBD	TBD	4,812	7,500,000		0	7,504,812	200,000	550,481
J1260	27,900,000	TBD	TBD	TBD	19,471,802	15,000,000		0	62,371,802	656,000	5,581,180
J1261	29,396,448	TBD	TBD	TBD	9,095,800	15,000,000		2,891,245	56,383,494	360,000	5,278,349
J1267	7,499,846	TBD	TBD	TBD	7,985,827	15,000,000		545,320	31,030,994	540,000	2,563,099
J1277	41,500,000	TBD	TBD	TBD	27,427,696	4,000,000		0	72,927,696	520,000	6,772,770
J1278	20,200,000	TBD	TBD	TBD	32,502,852	15,000,000		0	67,702,852	400,000	6,370,285
J1279	0	TBD	TBD	TBD	21,122,235	15,000,000		0	36,122,235	400,000	3,212,224
J1280	32,164,715	TBD	TBD	TBD	11,732,853	15,000,000		4,937,699	63,835,266	400,000	5,983,527
J1281	0	TBD	TBD	TBD	6,286,548	15,000,000		0	21,286,548	300,000	1,828,655
J1282	25,804,613	TBD	TBD	TBD	58,074,904	12,700,000		0	96,579,517	800,000	8,857,952
J1286	33,067,294	TBD	TBD	TBD	31,061,586	1,655,000		0	65,783,880	600,000	5,978,388
J1287	0	TBD	TBD	TBD	995,184	15,000,000		0	15,995,184	200,000	1,399,518
J1290	5,511,216	TBD	TBD	TBD	5,176,931	1,655,000		0	12,343,147	100,000	1,134,315
J1293	0	TBD	TBD	TBD	10,289,791	500,000		0	10,789,791	100,000	978,979
J1294	0	TBD	TBD	TBD	7,275,225	15,000,000		0	22,275,225	396,000	1,831,523

Project	ERIS Network Upgrades (\$)				NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrades (\$)	Total Network Upgrade Cost for Milestone Calculation (\$)	M2 (\$)	M3 (\$)
	Thermal	Stability	Short Circuit	Affected System	Deliverability	TO Network Upgrades	TO – Owned Direct Assigned				
a	b	c	d	e	f	g	h	i	j = b+c+d+f+g+i	\$4,000/MW	10% of (j) from Phase I-M2
J1347	0	TBD	TBD	TBD	10,223,222	15,000,000		0	25,223,222	700,000	1,822,322
J1351	35,300,000	TBD	TBD	TBD	9,751,504	15,000,000		0	60,051,504	480,000	5,525,150
J1361	0	TBD	TBD	TBD	4,799	4,000,000		0	4,004,799	200,000	200,480
J1364	44,594,614	TBD	TBD	TBD	3,880,870	15,000,000		283,979	63,759,463	600,000	5,775,946
J1367	0	TBD	TBD	TBD	9,766,226	3,270,000		0	13,036,226	380,000	923,623
J1368	0	TBD	TBD	TBD	54,682,705	7,500,000		0	62,182,705	800,000	5,418,271
J1372	0	TBD	TBD	TBD	27,341,353	7,500,000		0	34,841,353	400,000	3,084,135
J1373	7,919,047	TBD	TBD	TBD	2,056,237	15,000,000		0	24,975,284	380,000	2,117,528
J1380	25,294,551	TBD	TBD	TBD	54,446,330	12,700,000		0	92,440,882	800,000	8,444,088
J1384	9,928,390	TBD	TBD	TBD	58,420,942	12,700,000		0	81,049,332	800,000	7,304,933
J1400	44,677,595	TBD	TBD	TBD	3,880,870	15,000,000		332,488	63,890,953	600,000	5,789,095
J1402	0	TBD	TBD	TBD	0	0		0	0	0	0
J1404	87,000,000	TBD	TBD	TBD	18,392,239	10,000,000		0	115,392,239	668,000	10,871,224
J1414	6,900,000	TBD	TBD	TBD	2,381,553	7,500,000		0	16,781,553	600,000	1,078,155
J1415	0	TBD	TBD	TBD	4,101,353	4,000,000		0	8,101,353	1,000,000	0
J1421	12,500,000	TBD	TBD	TBD	32,246	15,000,000		0	27,532,246	1,200,000	1,553,225
J1424	25,294,493	TBD	TBD	TBD	54,402,061	12,700,000		0	92,396,555	800,000	8,439,655
J1425	0	TBD	TBD	TBD	2,124,047	0		0	2,124,047	840,000	0
J1431	0	TBD	TBD	TBD	41,820,343	0		0	41,820,343	800,000	3,382,034
J1434	10,858,216	TBD	TBD	TBD	7,857,037	4,500,000		0	23,215,252	400,000	1,921,525
J1436	0	TBD	TBD	TBD	3,576,193	0		0	3,576,193	200,000	157,619
J1437	0	TBD	TBD	TBD	1,515,529	2,500,000		0	4,015,529	720,000	0
J1439	0	TBD	TBD	TBD	5,774,950	7,500,000		0	13,274,950	160,000	1,167,495
J1440	0	TBD	TBD	TBD	23,099,798	7,500,000		0	30,599,798	640,000	2,419,980
J1441	0	TBD	TBD	TBD	40,251,424	4,500,000		0	44,751,424	800,000	3,675,142
J1442	0	TBD	TBD	TBD	48,283,312	15,000,000		0	63,283,312	660,000	5,668,331
J1452	0	TBD	TBD	TBD	20,991,324	10,120,000		0	31,111,324	800,000	2,311,132
J1455	0	TBD	TBD	TBD	15,454,645	10,120,000		0	25,574,645	600,000	1,957,465
J1458	0	TBD	TBD	TBD	31,392,719	15,000,000		0	46,392,719	800,000	3,839,272
J1462	2,900,000	TBD	TBD	TBD	48,412,638	15,000,000		0	66,312,638	600,000	6,031,264
J1463	0	TBD	TBD	TBD	30,843,670	4,500,000		0	35,343,670	600,000	2,934,367
J1465	0	TBD	TBD	TBD	22,303,042	15,000,000		0	37,303,042	600,000	3,130,304
J1469	62,227,945	TBD	TBD	TBD	5,178,152	15,000,000		284,585	82,690,682	800,000	7,469,068

Analyses performed demonstrate the following transmission facilities are required to reliably interconnect this group of generators to the transmission system. ERIS Network Upgrades and NRIS Network Upgrades are shown in Table 3. Shared Network Upgrades are shown in Table 4.



Table 3: ERIS & NRIS Upgrades (Planning level cost estimates)

Network Upgrade	TO	GI projects requiring upgrade for ERIS	GI projects requiring upgrade for NRIS	Cost of solution (\$)
Bayou Labutte - Pt. Pleasant 230 kV Line	EES	J1246		\$ 14,500,000
Bayou Labutte 800 MVA Autotransformer #2	EES	J1246		\$ 30,000,000
Brinkley East - Devalls Bluff 115 kV Upgrade	EES	J1261 J1280 J1373 J1434		\$ 35,100,000
Coly - Moler 230 kV Upgrade	EES	J1421		\$ 12,500,000
Delta - Drew 115 kV Upgrade	EES	J1404		\$ 23,700,000
Eunice - Manuel Reconductor	Cleco	J1286 J1290		\$ 22,184,000
Gordon - Magnolia Ames 138 kV Upgrade	EES	J1277		\$ 12,100,000
J1217 POI - Westfork Reconductor	Cleco	J1205 J1217 J1220 J1236 J1238 J1286 J1290 J1384		\$ 20,448,000
J1260 POI - Dumas 115 kV Upgrade	EES	J1260		\$ 7,700,000
J1260 POI - Reed 115 kV Upgrade	EES	J1260		\$ 7,700,000
J1260 POI - Reed 115 kV New Line #2	EES	J1260		\$ 12,500,000
J1261 POI - Brinkley East 115 kV Upgrade	EES	J1215 J1261 J1280		\$ 13,100,000
J1278 POI - L111T411 138 kV Upgrade	EES	J1278		\$ 20,200,000
J1280 POI - Clarendon 115 kV Upgrade	EES	J1261 J1280		\$ 6,100,000
J1282 POI - Penton Reconductor	Cleco	J1205 J1220 J1238 J1282 J1380 J1424		\$ 17,248,000
J1351 POI - Delhi 115 kV Upgrade	EES	J1351		\$ 14,900,000
J1351 POI - Tallulah 115 kV Upgrade	EES	J1351		\$ 20,400,000
J1400 POI - J1469 POI 161 kV Upgrade, J1400 POI - J1469 POI 161 kV New 2nd Line	EES	J1364 J1400 J1469		\$ 4,000,000
J1462 POI - Scanlan 138 kV Upgrade	EES	J1462		\$ 2,900,000
Magnolia Ames - Pintail 138 kV Upgrade	EES	J1277		\$ 7,600,000
Morrow - Purvis Bulk 161 kV New Line, Upgrade Purvis Bulk Substation	CE	J1249		\$ 5,847,500
Moss Bluff - J1424 POI New Line	Cleco	J1282 J1380 J1424		\$ 60,950,000
Mt. Home - Midway Jordan 161 kV Upgrade	EES	J1414		\$ 6,900,000
Pintail 138 kV Redundant Breaker	EES	J1277		\$ 10,100,000
Raywood - Pintail 138 kV Upgrade	EES	J1277		\$ 11,700,000
Richard - Eunice Reconductor	Cleco	J1286 J1290		\$ 13,700,000
Ruleville - Drew 115 kV New Line	EES	J1404		\$ 42,800,000
Ruleville - Schlater 115 kV Upgrade	EES	J1404		\$ 20,500,000
Stuttgart Industrial - Ulm 115 kV Upgrade	EES	J1261 J1280		\$ 7,500,000
Stuttgart North - Humphrey 115 kV Upgrade	EES	J1215		\$ 10,300,000
Stuttgart North - Stuttgart Ricuskey 115 kV Upgrade	EES	J1215 J1245		\$ 4,000,000
Ulm - Clarendon 115 kV Upgrade	EES	J1261 J1280		\$ 20,100,000
West Memphis - J1469 POI - Sans Souci 500 kV New Line, Two J1469 POI 500/161 kV Autotransformers	EES	J1267 J1364 J1400 J1469		\$ 155,000,000
Westfork - Wells Reconductor	Cleco	J1205 J1217 J1220 J1236 J1238 J1384		\$ 14,764,000
2nd Clarksdale Transformer	Cleco		J1215 J1245 J1262 J1280 J1293 J1434	\$6,000,000

Network Upgrade	TO	GI projects requiring upgrade for ERS	GI projects requiring upgrade for NRIS	Cost of solution (\$)
Alzheimer - Woodward 115 kV Upgrade	EES		J1215 J1245 J1261 J1280 J1293 J1434	\$22,000,000
Andrus - Refuge 115 kV Upgrade	EES		J1260 J1458	\$-
Andrus 230/115 kV 2nd Autotransformer	EES		J1260 J1281 J1458	\$18,000,000
Bayou Labutte - Willow Glen 500 kV Upgrade	EES		J1205 J1217 J1220 J1236 J1238 J1246 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462 J1465	\$26,700,000
Big Cajun - Fancy Point 500 kV Upgrade	EES		J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	\$29,600,000
Big Cajun - Horseshoe 230 kV Upgrade	EES		J1368 J1372 J1463	\$78,100,000
Bogalusa 500/230 kV 2nd Autotransformer, Add breakers at Bogalusa 500 kV	EES		J1239 J1260 J1281 J1294 J1347 J1351 J1404 J1436 J1458	\$66,500,000
Cocodrie - Big Cajun New Line	Cleco		J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1368 J1372 J1380 J1384 J1424 J1442 J1452 J1455 J1462 J1463 J1465	\$198,750,000
Cocodrie - Ville Plate Reconductor	Cleco		J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	\$22,600,000
Delta - Cleveland 115 kV Upgrade	EES		J1404 J1439 J1440	\$6,300,000
El Dorado - Mt. Olive 500 kV Upgrade	EES		J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1434 J1236 J1437 J1469	\$2,000,000
Frisco - Convent 230 kV Upgrade	EES		J1431 J1441	\$46,000,000
Greenville - SE Greenville 115 kV Upgrade	EES		J1458	\$5,300,000
Gypsy - Ironman 230 kV Upgrade	EES		J1431 J1441	\$500,000
Horn Lake - Freeport 161 kV Upgrade*	EES		J1215 J1245 J1261 J1280 J1287 J1293 J1404 J1434 J1439 J1440	\$10,100,000
Isola SS - Belzoni 115 kV Upgrade	EES		J1294 J1458	\$500,000
J1007 POI - J1469 POI 161 kV Upgrade	EES		J1250 J1267 J1414	\$1,900,000
J1060 POI - West Memphis 500 kV Upgrade	EES		J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	\$1,000,000
J1278 POI - Moscow 138 kV Upgrade	EES		J1278 J1442	\$7,000,000
J1384 POI - J1217 POI Reconductor	Cleco		J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1465	\$4,050,000
J1424 POI - Rork 230 kV New Line	Cleco		J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1462 J1465	\$162,000,000
J1440 POI - Tunica 115 kV Upgrade	EES		J1294 J1404 J1439 J1440	\$28,600,000
J552 POI - Brinkley East 115 kV 2nd New Line	EES		J1293	\$8,200,000
J620 POI - Macon Lake 115 kV Upgrade	EES		J1260 J1281 J1425	\$7,900,000

Network Upgrade	TO	GI projects requiring upgrade for ERS	GI projects requiring upgrade for NRIS	Cost of solution (\$)
LTCR for Purvis Bulk 230/161 kV Transformers	CE		J1249	\$-
Manena - Sadie 230 kV Upgrade	EES		J1205 J1282 J1380 J1424	\$9,600,000
Moscow - Corrigan 138 kV Upgrade	EES		J1278 J1442	\$10,200,000
Pelican Road - Jacinto 138 kV Upgrade	EES		J1278 J1442	\$13,700,000
Perryville - Baxter Wilson 500 kV Upgrade	EES		J1215 J1239 J1245 J1258 J1260 J1261 J1280 J1281 J1373 J1415 J1425 J1436 J1437 J1452	\$1,000,000
Ray Braswell - Franklin 500 kV Upgrade	EES		J1215 J1239 J1245 J1260 J1261 J1280 J1281 J1287 J1293 J1294 J1347 J1351 J1373 J1404 J1425 J1434 J1436 J1439 J1440 J1458	\$3,000,000
Refuge - Greenville Industrial 115 kV Upgrade	EES		J1260 J1458	\$-
Rork - Wells 230 kV New Line	Cleco		J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1462 J1465	\$55,000,000
Sabine - Mud Lake 230 kV Upgrade	EES		J1277 J1278 J1279 J1442	\$400,000
Sans Souci - Driver 500 kV Upgrade	EES		J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	\$3,300,000
Scanlan - Bosco 138 kV Upgrade	EES		J1286 J1290 J1462	\$9,900,000
Tezcuco - Frisco #1 230 kV Upgrade	EES		J1431 J1441	\$2,000,000
Tezcuco - Frisco #2 230 kV Upgrade	EES		J1431 J1441	\$2,000,000
Wabaseka - Humphrey 115 kV Upgrade	EES		J1215 J1245 J1261 J1280 J1293 J1434	\$10,300,000
Wabaseka - Wabaseka West 115 kV Upgrade	EES		J1215 J1245 J1261 J1280 J1293 J1434	\$4,000,000
Waterloo - Big Cajun 230 kV Upgrade	EES		J1463	\$3,100,000
Webre - Bayou Labutte 500 kV Upgrade	EES		J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	\$89,900,000
W Memphis - Birmingham Steel 500 kV Upgrade*	EES		J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	\$31,500,000
Wabaseka West - Alzheimer 115 kV Upgrade	EES		J1215 J1245 J1261 J1280 J1293 J1434	\$6,800,000
East Leesville - Rodemacher Reconductor	Cleco		J1217 J1220 J1236 J1238 J1384	\$67,408,000
Kuhn Rd - Ebony 161 kV Upgrade	EES		J1267	\$4,000,000
J1282 POI - J1424 POI Reconductor	Cleco		J1205 J1217 J1220 J1236 J1238 J1282 J1380 J1384 J1424	\$4,800,000
Rapides - Forest Hill Reconductor	Cleco		J1220 J1238 J1452	\$42,768,000
Rilla - Riverton 115 kV Line Upgrade	EES		J1239	\$33,700,000
Waterloo - J1368 POI 230 kV Upgrade	EES		J1368 J1372	\$7,000,000
Wynne South - Wynne East 161 kV Upgrade	EES		J1261 J1280 J1293 J1373 J1434	\$6,100,000
J1220 POI - Rapides Reconductor	Cleco		J1220 J1238 J1452	\$10,000,000
Waterford - Tezcuco 230 kV Upgrade	EES		J1367 J1431 J1441 J1455	\$38,500,000

*A portion of these Network Upgrades involve Affected Systems that are not the TO listed and could see cost increases when that analysis is



performed in Phase 2.

Table 4: Shared Network Upgrades (Planning level cost estimates)

Shared Network Upgrade	NU Study Cycle	Higher queued projects associated with SNU	Study projects associated with SNU	Cost of solution (\$)
Stuttgart Ricuskey – Stuttgart Industrial 115 kV Reconductor	DPP-2014-AUG	J348	J1215 J1261 J1280	11,400,000
Kuhn Rd - Ebony 161 kV Reconductor	DPP-2017-AUG	J934 J1007 J1125	J1267 J1364 J1400 J1469	4,900,000
Bogalusa – Franklin 500 kV Upgrade	DPP-2017-AUG	J830 J868 J908	J1249	500,000

Note:

- 1) Details pertaining to upgrades, costs, and the execution plan for interconnection of the generating facility at the POI will be documented in the Facility Study for Interconnecting Generator.
- 2) Facilities that have been included as base case assumptions and the level of interconnection service that would be conditional upon these facilities being in service will be documented in the GIA (Generator Interconnection Agreement) for each respective GI request successfully achieving GIA execution.
- 3) Analysis performed shows there are three projects for Shared Network Upgrade cost allocation.
- 4) A detailed summary of the estimated cost of Network Upgrades on a per project basis can be found in Appendix B – Network Upgrade Table Per Project (CEII).

2. FERC Order 827 Compliance Review

The Final Rule of FERC Order 827 “Reactive Power Requirements for Non-Synchronous Generation”, which was issued June 16, 2016, states that “Under this Final Rule, newly interconnecting non-synchronous generators that have not yet executed a Facilities Study Agreement as of the effective date of this Final Rule will be required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.” As such, this Final Rule applies to all wind and solar projects included in the DPP 2019 Cycle 1 South study cycle.

In this study, the power factor at the high-side of the generator substation for each inverter-based project was calculated and reviewed. The study method is to set Qgen of each study project at its Qmax, solve the case, then record the P and Q injection on the high side of the generator substation to calculate the lagging power factor (injecting VAR to the system). The same process is then repeated by setting Qgen at Qmin to calculate the leading power factor (absorbing VAR from the system).

The results show that all projects meet the requirement to maintain 0.95 leading power factor, however, 32 projects do not meet the requirement to provide reactive power capability corresponding to 0.95 lagging power factor, as highlighted in red below in Table 5. Additional reactive support will be needed for these projects to meet the FERC requirement on reactive power capability prior to the completion of their GIA.

Table 5: FERC Order 827 Review Results

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensation	VAR Injection			VAR Absorption			Meet FERC Order 827 Requirement?	Add'l VAR Needed (MVar)
				P (MW)	Q (MVar)	Lagging p.f (pu)	P (MW)	Q (MVar)	Leading p.f. (pu)		
J1205	125	±41.0900	1 x 18 MVar Cap	123.4	37.9	0.9559	123.2	-63.5	-0.8889	No	2.7
J1215	200	±65.74	2 x 3 MVar Cap	198.3	40.3	0.9800	197.6	-112.9	-0.8683	No	24.9
J1217	200	±65.74	2 x 3 MVar Cap	198.3	39.4	0.9808	197.7	-110.3	-0.8733	No	25.8
J1219	100	±32.87	1 x 3 MVar Cap	99.1	18.8	0.9825	98.8	-56.4	-0.8685	No	13.8
J1220	200	±65.736	2 x 3 MVar Cap	198.2	41.2	0.9791	197.9	-106.5	-0.8806	No	23.9
J1223	150	±49.3000	1 x 4 MVar Cap	149.1	47	0.9537	148.8	-59.3	-0.9289	No	2.0
J1236	50	±24.2	1 x 2 MVar Cap	49.6	17.5	0.9430	49.4	-37.1	-0.7996	Yes	
J1238	50	±24.2200	1 x 2 MVar Cap	49.6	18.1	0.9394	49.4	-36.1	-0.8074	Yes	
J1239	100	±50	None	99	39.2	0.9298	98.6	-66.6	-0.8287	Yes	
J1245	50	±16.4300	1 x 2 MVar Cap	49.6	12.6	0.9692	49.5	-25.1	-0.8919	No	3.7
J1250	35	±11.5000	None	34.8	10.1	0.9604	34.7	-13.5	-0.9320	No	1.3
J1257	20	±9.6900	None	19.8	6.2	0.9543	19.8	-19.7	-0.7089	No	0.3
J1258	50	±16.4300	1 x 2 MVar Cap	49.3	14.8	0.9578	49.2	-22.5	-0.9094	No	1.4
J1259	50	±24.2200	1 x 2 MVar Cap	49.6	17.7	0.9418	49.3	-38.1	-0.7912	Yes	
J1260	164	±79.4300	1 x 37 MVar	161	100.5	0.8483	158	-136.2	-0.7574	Yes	

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensation	VAR Injection			VAR Absorption			Meet FERC Order 827 Require ment?	Add'l VAR Needed (MVar)
				P (MW)	Q (MVar)	Lagging p.f (pu)	P (MW)	Q (MVar)	Leading p.f. (pu)		
			Cap								
J1261	90	±43.5890	1 x 19 MVar Cap	88.5	52.2	0.8613	87.7	-66.6	-0.7964	Yes	
J1267	135	±43.7450	2 x 12 MVar Cap 1 x 9 MVar Cap	135.5	59.9	0.9146	134.5	-79.7	-0.8603	Yes	
J1277	130	±42.7300	1 x 4 MVar Cap	128.9	24.1	0.9830	128.4	-74.8	-0.8641	No	18.3
J1278	102.6	±63.5900	None	99.9	47.2	0.9042	98.3	-103.3	-0.6894	Yes	
J1279	102.6	±63.5900	None	99.9	47.2	0.9042	98.3	-103.3	-0.6894	Yes	
J1280	102.6	±63.5900	None	100	47.7	0.9026	99	-93.8	-0.7259	Yes	
J1281	75	±50.2000	None	74.9	38.2	0.8908	74.1	-74.5	-0.7052	Yes	
J1282	200	±133.8600	None	199.3	98.7	0.8961	195.1	-214.7	-0.6725	Yes	
J1286	150	±49.3000	1 x 4 MVar Cap	147.5	46.7	0.9534	146.7	-68.5	-0.9061	No	1.8
J1287	50	±27.3600	None	50	21.1	0.9213	49.6	-41.1	-0.7700	Yes	
J1290	25	±8.2200	None	24.8	4.3	0.9853	24.8	-13.2	-0.8827	No	3.9
J1293	25	±8.2200	None	24.8	4.3	0.9853	24.8	-13.3	-0.8813	No	3.9
J1294	99	±40.8000	None	99	25.5	0.9684	97.1	-67.1	-0.8227	No	7.0
J1347	175	±101.3700	None	172.8	83.2	0.9010	169.9	-155.4	-0.7379	Yes	
J1351	120	±69.3600	None	118.2	55.7	0.9046	116.2	-108.5	-0.7309	Yes	
J1361	50	±17.3900	None	52.1	8.3	0.9875	51.8	-29.5	-0.8690	No	8.8
J1364	150	±87.1400	None	148.5	75.2	0.8921	147.5	-108.2	-0.8063	Yes	
J1367	95	±32.63	None	94.1	20.6	0.9769	93.8	-47.9	-0.8906	No	10.3
J1368	200	±65.7000	1 x 10 MVar Cap	199.1	27.6	0.9905	197.2	-142.6	-0.8103	No	37.8
J1372	100	±38.5950	1 x 12 MVar Cap	100.6	9.3	0.9958	98.2	-62.1	-0.8452	No	23.8
J1373	95	±45.6700	1 x 6 MVar Cap	95	38.2	0.9278	94.4	-73.2	-0.7903	Yes	
J1380	200	±94.0100	None	196.7	62.9	0.9525	195.1	-141	-0.8105	No	1.8
J1384	200	±67.0500	None	199.9	20.6	0.9947	197.8	-140	-0.8162	No	45.1
J1400	150	±49.7000	None	148.7	50.4	0.9471	148.4	-53.1	-0.9415	Yes	
J1402 (w J919)	180	±79.25	1 x 42 MVar Cap	180	95.1	0.8842	179	-131.7	-0.8055	Yes	
J1404	167	±80.8800	1 x 5 MVar Cap	163.8	71.6	0.9163	162.2	-103.5	-0.8430	Yes	
J1414	150	±49.3	1 x 4 MVar Cap	147.5	47.2	0.9524	146.3	-73.5	-0.8936	No	1.3
J1415	250	±82.18	2 x 4 MVar Cap	245.3	61.5	0.9700	244.1	-124.7	-0.8905	No	19.1
J1421	300	±50.7400	2 x 38 MVar Cap	295.2	135.8	0.9085	293.5	-139.8	-0.9028	Yes	
J1424	200	±65.7400	1 x 14 MVar Cap	197.8	46.3	0.9737	197.1	-110.9	-0.8715	No	18.7
J1431	200	±96.86	1 x 30	193.6	90.4	0.9061	188.1	-181.1	-0.7204	Yes	

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensation	VAR Injection			VAR Absorption			Meet FERC Order 827 Requirement?	Add'l VAR Needed (MVar)
				P (MW)	Q (MVar)	Lagging p.f (pu)	P (MW)	Q (MVar)	Leading p.f. (pu)		
J1434	100	±32.87	2 x 7 MVar	98.9	32.7	0.9494	98.6	-53.9	-0.8775	Yes	
J1436	50	±16.53	3 x 7.5	49.9	37	0.8033	49.8	-20.9	-0.9221	Yes	
J1437	180	±87.38	4 x 10.5 MVar Cap	175.4	115.2	0.8358	151.5	-162.4	-0.4227	Yes	
J1439	40	±13.15	1 x 2 MVar Cap	39.7	8.8	0.9763	39.6	-22	-0.8742	No	4.2
J1440	160	±52.59	1 x 5 MVar Cap	158.7	32.2	0.9800	158	-93.7	-0.8601	No	20.0
J1441	200	±71	2 x 14 MVar Cap	200.1	66	0.9497	199.1	-121.1	-0.8544	Yes	
J1442	165	±62.45	1 x 6 MVar Cap	163.8	44.8	0.9646	163.2	-99.3	-0.8543	No	9.0
J1452	200	±32.87	2 x 3 MVar Cap	198.3	40.2	0.9801	197.6	-106.7	-0.8799	No	25.0
J1455	150	±49.3	2 x 11 MVar Cap	148.4	50.2	0.9473	147.8	-82.1	-0.8742	Yes	
J1458	200	±65.74	2 x 14 MVar Cap	197.4	63.7	0.9517	197	-112.9	-0.8676	No	1.2
J1462	150	±49.3	2 x 11 MVar Cap	148.4	49.8	0.9480	147.8	-82.7	-0.8727	Yes	
J1463	150	±52.4	None	147.4	23.5	0.9875	146.3	-93.8	-0.8418	No	24.9
J1465	150	±49.3	2 x 14 MVar Cap	148.6	61.3	0.9244	148	-76.2	-0.8891	Yes	
J1469	200	±65.84	2 x 14 MVar Cap	198	63.6	0.9521	197.3	-108.7	-0.8759	No	1.5
J1205	125	±41.0900	1 x 18 MVar Cap	123.4	37.9	0.9559	123.2	-63.5	-0.8889	No	2.7

3. Model Development and Study Assumptions

3.1. Base Case Models

The origin of the DPP 2019 Cycle 1 South models is the MTEP19 models with the Bench Cases including all pre-queued projects and associated Network Upgrades, while the Study Cases contain all of the interconnection requests in DPP 2019 Cycle 1 South Phase I, in addition to all the facilities in the Bench Cases.

- Bench Cases
 - DPP19C1-PhaseI-2024SUM-Bench_v1
- Study Cases
 - DPP19C1-PhaseI-2024SUM-Study-Charging_v1
 - DPP19C1-PhaseI-2024SUM-Study-Discharging_v1

3.2. Monitored Elements

Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal rating (PSS®E Rating A), and for NERC category P1-P7 conditions branches were monitored for emergency rating (PSS®E Rating B). Voltage limits were specified for system intact and contingent conditions as per applicable Transmission Owner Planning Criteria.

3.3. Contingencies

The following contingencies were considered in the steady state analysis:

- 1) NERC Category P0 (system intact -- no contingencies)
- 2) NERC Category P1 contingencies
 - a. Single element outages, at buses with a nominal voltage of 68 kV and above
 - b. Multiple element NERC Category P1 contingencies
- 3) NERC Category P2-P7 contingencies
- 4) For all the contingencies and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

3.4. Study Methodology

Non-linear (AC) contingency analysis was performed on the benchmark and study cases, and the incremental impact of the DPP 2019 Cycle 1 South generating facilities was evaluated by comparing the steady state performance of the transmission system in the Bench and Study Cases. Analyses used PSS®E version 33.11.0 and TARA version 1902.

3.5. Performance Criteria

A branch is considered a thermal constraint if the following conditions are met:

- 1) The generator has a larger than twenty percent (20%) sensitivity factor on the overloaded facilities under post-contingent condition (see NERC TPL) or five percent (5%) sensitivity factor under system-intact condition, or
- 2) The overloaded facility or the overload-causing contingency is at generator's outlet, or
- 3) The megawatt impact due to the generator is greater than or equal to twenty percent (20%) of the applicable rating (normal or emergency) of the overloaded facility, or
- 4) For any other constrained facility, where none of the Study Generators meet one of the above criteria, however, the cumulative MW impact of the group of study generators is greater than twenty percent (20%) of the rating of the facility, then only those study generators whose individual MW impact is greater than five percent (5%) of the rating of the facility and has DF greater than five percent (5%) will be responsible for mitigating the cumulative MW impact constraint, or
- 5) Impacts on Affected Systems would be classified as injection constraints based on the Affected Systems' criteria, or
- 6) Any other applicable Transmission Owner Local Planning Criteria are met.

A bus is considered a voltage constraint if both of the following conditions are met:

- 1) The bus voltage is outside of the applicable normal or emergency limits for the post change case, and
- 2) The change in bus voltage is greater than 0.01 per unit

All generators must mitigate thermal injection constraints and voltage constraints in order to obtain any type of Interconnection Service. Further, all generators requesting NRIS must mitigate constraints found by using the Deliverability algorithm to meet the system performance criteria for NERC category P1 events, if DFAX due to the study generator is equal to or greater than 5%.

4. Thermal Analysis

The thermal analysis results for 2024 Summer show generator projects J1205, J1215, J1217, J1220, J1236, J1238, J1245, J1246, J1249, J1260, J1261, J1267, J1277, J1278, J1280, J1282, J1286, J1290, J1351, J1364, J1373, J1380, J1384, J1400, J1404, J1414, J1421, J1424, J1434, J1462, and J1469 causing constraints. The details pertaining to the thermal analysis can be found in Appendix C – MISO ERIS Analysis (CEII).

4.1. J1205



Thermal analysis determined that this generator contributes to the following constraints:

1. J1282 POI – Penton 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the J1282 POI – Penton line. The cost estimate is \$17.248 million, with the project allocated 3.9% of the cost.

2. J1217 POI – Westfork 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the J1217 POI – Westfork line. The cost estimate is \$20.448 million, with the project allocated 5.4% of the cost.

3. Westfork – Wells 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the Westfork – Wells line. The cost estimate is \$14.764 million, with the project allocated 6.1% of the cost.

4. Nelson – Penton 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is a new line from Moss Bluff – J1424 POI. The cost estimate is \$60.95 million, with the project allocated 1.2% of the cost.

4.2. J1215

Thermal analysis determined that this generator contributes to the following constraints:

1. J1261 POI – Brinkley East 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the J1261 POI – Brinkley East line. The cost estimate is \$13.1 million, with the project allocated 11.9% of the cost.

2. Stuttgart North – Stuttgart Ricuskey 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the Stuttgart North – Stuttgart Ricuskey line. The cost estimate is \$4 million, with the project allocated 80% of the cost.

3. Stuttgart North – Humphrey 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the Stuttgart North – Humphrey line. The cost estimate is \$10.3 million, with the project allocated 100% of the cost.

4.3. J1217

Thermal analysis determined that this generator contributes to the following constraints:

1. J1217 – Westfork 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the J1217 – Westfork line. The cost estimate is \$20.448 million, with the project allocated 28.3% of the cost.

2. Westfork – Wells 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the Westfork – Wells line. The cost estimate is \$14.764 million, with the project allocated 33.4% of the cost.

4.4. J1220

Thermal analysis determined that this generator contributes to the following constraints:

1. J1282 POI – Penton 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the J1282 POI – Penton line. The cost estimate is \$17.248 million, with the project allocated 1.9% of the cost.

2. J1217 POI – Westfork 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the J1217 POI - Westfork line. The cost estimate is \$20.448 million, with the project allocated 15.8% of the cost.

3. Westfork – Wells 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the Westfork - Wells line. The cost estimate is \$14.764 million, with the project allocated 17% of the cost.

4.5. J1236

Thermal analysis determined that this generator contributes to the following constraints:

1. J1217 – Westfork 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the J1217 – Westfork line. The cost estimate is \$20.448 million, with the project allocated 7.1% of the cost.

2. Westfork – Wells 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the Westfork – Wells line. The cost estimate is \$14.764 million, with the project allocated 8.3% of the cost.

4.6. J1238

Thermal analysis determined that this generator contributes to the following constraints:

1. J1282 POI – Penton 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the J1282 POI – Penton line. The cost estimate is \$17.248 million, with the project allocated 0.5% of the cost.

2. J1217 POI – Westfork 230 kV



Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the J1217 POI - Westfork line. The cost estimate is \$20.448 million, with the project allocated 4.0% of the cost.

3. Westfork – Wells 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the Westfork - Wells line. The cost estimate is \$14.764 million, with the project allocated 4.3% of the cost.

4.7. J1245

Thermal analysis determined that this generator contributes to the following constraints:

1. Stuttgart North – Stuttgart Ricuskey 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the Stuttgart North – Stuttgart Ricuskey line. The cost estimate is \$4 million, with the project allocated 20% of the cost.

4.8. J1246

Thermal analysis determined that this generator contributes to the following constraints:

1. Bayou Labutte 800 MVA Autotransformer

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to add a second Bayou Labutte 800 MVA autotransformer. The cost estimate is \$30 million, with the project allocated 100% of the cost.

2. Iberville – Bayou Labutte 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is a new Bayou Labutte – Pt. Pleasant 230 kV line. The cost estimate is \$14.5 million, with the project allocated 100% of the cost.

4.9. J1249

Thermal analysis determined that this generator contributes to the following constraints:

1. Purvis Bulk – Oak Grove 69 kV
2. Purvis Bulk – Lumber 69 kV
3. Oak Grove 161/69 kV Transformer

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraints. The planned upgrade is to add a new line from Morrow – Purvis Bulk and upgrade the Purvis Bulk substation. The cost estimate is \$5.847 million, with the project allocated 100% of the cost.

4.10. J1260

Thermal analysis determined that this generator contributes to the following constraints:

1. J1260 POI – Reed 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the J1260 POI – Reed line. The cost estimate is \$7.7 million, with the project allocated 100% of the cost.

2. Dumas – Dumas North 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to add a new line from J1260 POI - Reed. The cost estimate is \$12.5 million, with the project allocated 100% of the cost.

3. J1260 POI – Dumas 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the J1260 POI – Dumas line. The cost estimate is \$7.7 million, with the project allocated 100% of the cost.

4.11. J1261

Thermal analysis determined that this generator contributes to the following constraints:

1. Brinkley East – Devalls Bluff 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the Brinkley East – Devalls Bluff line. The cost estimate is \$35.1 million, with the project allocated 23.1% of the cost.

2. J1261 POI – Brinkley East 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the J1261 – Brinkley East line. The cost estimate is \$13.1 million, with the project allocated 41.5% of the cost.

3. J1280 POI – Clarendon 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the J1280 POI – Clarendon line. The cost estimate is \$6.1 million, with the project allocated 47.1% of the cost.

4. Ulm – Clarendon 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the Ulm – Clarendon line. The cost estimate is \$20.1 million, with the project allocated 47.1% of the cost.

5. Stuttgart Industrial – Ulm 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the Stuttgart Industrial – Ulm line. The cost estimate is \$7.5 million, with the project allocated 47.1% of the cost.

4.12. J1267

Thermal analysis determined that this generator contributes to the following constraints:

1. Kuhn Rd – Ebony 161 kV
2. Ebony – WM Lehigh-West 161 kV
3. Ebony – WM Lehigh-East 161 kV
4. W Memphis – WM Lehigh-West 161 kV
5. W Memphis – WM Polk Ave 161 kV



6. W Memphis 500/161 kV Transformer
7. Marked Tree – Harrisburg 161 kV
8. Harrisburg – Trumann 161 kV
9. WM Lehigh-East – WM Polk Ave 161 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraints. The planned upgrade is a new line from West Memphis – J1469 POI – Sans Souci 500 kV and two 500/161 kV autotransformers at the J1469 POI. The cost estimate is \$155 million, with the project allocated 4.8% of the cost.

4.13. J1277

Thermal analysis determined that this generator contributes to the following constraints:

1. Raywood – Pintail 138 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the Raywood - Pintail line. The cost estimate is \$11.7 million, with the project allocated 100% of the cost.

2. Gordon – Magnolia Ames 138 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the Gordon – Magnolia Ames line. The cost estimate is \$12.1 million, with the project allocated 100% of the cost.

3. Magnolia Ames – Pintail 138 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the Magnolia Ames - Pintail line. The cost estimate is \$7.6 million, with the project allocated 100% of the cost.

4. Pintail – W Bayou 138 kV
5. Shiloh – Barkers Corner 138 kV
6. Barkers Corner – Bayshore 138 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraints. The planned upgrade is adding a redundant breaker at Pintail. The cost estimate is \$10.1 million, with the project allocated 100% of the cost.

4.14. J1278

Thermal analysis determined that this generator contributes to the following constraints:

1. J1278 POI – L111T411 138 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the J1278 POI – L11T411 line. The cost estimate is \$20.2 million, with the project allocated 100% of the cost.

4.15. J1280

Thermal analysis determined that this generator contributes to the following constraints:

1. Brinkley East – Devalls Bluff 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the



listed constraint. The planned upgrade is to upgrade the Brinkley East – Devalls Bluff line. The cost estimate is \$35.1 million, with the project allocated 23.4% of the cost.

2. J1261 POI – Brinkley East 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the J1261 POI – Brinkley East line. The cost estimate is \$13.1 million, with the project allocated 46.6% of the cost.

3. J1280 POI – Clarendon 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the J1280 POI – Clarendon line. The cost estimate is \$6.1 million, with the project allocated 52.9% of the cost.

4. Ulm – Clarendon 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the Ulm – Clarendon line. The cost estimate is \$20.1 million, with the project allocated 52.9% of the cost.

5. Stuttgart Industrial – Ulm 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the Stuttgart Industrial – Ulm line. The cost estimate is \$7.5 million, with the project allocated 52.9% of the cost.

4.16. J1282

Thermal analysis determined that this generator contributes to the following constraints:

1. J1282 POI – Penton 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the J1282 POI - Penton line. The cost estimate is \$17.248 million, with the project allocated 52.9% of the cost.

2. Cooper 230/138 kV Transformer
3. J1380 POI – Cooper 230 kV
4. Cooper – Deridder Mill 230 kV
5. Nelson – Penton 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is a new line from Moss Bluff – J1424 POI. The cost estimate is \$60.95 million, with the project allocated 33.4% of the cost.

4.17. J1286

Thermal analysis determined that this generator contributes to the following constraints:

1. J1217 POI – Westfork 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the J1217 POI – Westfork line. The cost estimate is \$20.448 million, with the project allocated 11.3% of the cost.



2. Eunice – Manuel 138 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the Eunice – Manuel line. The cost estimate is \$22.184 million, with the project allocated 85.7% of the cost.

3. Richard – Eunice 138 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the Richard - Eunice line. The cost estimate is \$13.7 million, with the project allocated 85.7% of the cost.

4.18. J1290

Thermal analysis determined that this generator contributes to the following constraints:

1. J1217 POI – Westfork 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the J1217 POI – Westfork line. The cost estimate is \$20.448 million, with the project allocated 1.9% of the cost.

2. Eunice – Manuel 138 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the Eunice – Manuel line. The cost estimate is \$22.184 million, with the project allocated 14.3% of the cost.

3. Richard – Eunice 138 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the Richard - Eunice line. The cost estimate is \$13.7 million, with the project allocated 14.3% of the cost.

4.19. J1351

Thermal analysis determined that this generator contributes to the following constraints:

1. J1351 POI – Delhi 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the J1351 POI - Delhi line. The cost estimate is \$14.9 million, with the project allocated 100% of the cost.

2. J1351 – Tallulah 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the J1351 POI - Tallulah line. The cost estimate is \$20.4 million, with the project allocated 100% of the cost.

4.20. J1364

Thermal analysis determined that this generator contributes to the following constraints:

1. J1400 POI – J1469 POI 161 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the



listed constraint. The planned upgrade is to upgrade the J1400 POI – J1469 POI line and construct a new J1400 POI – J1469 POI line. The cost estimate is \$4 million, with the project allocated 30% of the cost.

2. J1007 POI – J1469 POI 161 kV
3. J1007 POI – Marked Tree 161 kV
4. Parkin – Kuhn Rd 161 kV
5. Parkin – Twist 161 kV
6. Twist – Marked Tree 161 kV
7. Powerline Rd – Trumann W 161 kV
8. Ebony – WM Lehigh-West 161 kV
9. Ebony – WM Lehigh-East 161 kV
10. W Memphis – WM Lehigh-West 161 kV
11. W Memphis – WM Polk Ave 161 kV
12. W Memphis 500/161 kV Transformer
13. Marked Tree – Harrisburg 161 kV
14. Harrisburg – Trumann 161 kV
15. Trumann – Trumann West 161 kV
16. WM Lehigh-East – WM Polk Ave 161 kV
17. Ebony – Marion 161 kV
18. J1400 POI – Marion 161 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraints. The planned upgrade is a new line from West Memphis – J1469 POI – Sans Souci 500 kV and two 500/161 kV autotransformers at the J1469 POI. The cost estimate is \$155 million, with the project allocated 28% of the cost.

4.21. J1373

Thermal analysis determined that this generator contributes to the following constraints:

1. Brinkley East – Devalls Bluff 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the Brinkley East – Devalls Bluff line. The cost estimate is \$35.1 million, with the project allocated 22.6% of the cost.

4.22. J1380

Thermal analysis determined that this generator contributes to the following constraints:

1. J1282 POI – Penton 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the J1282 – Penton line. The cost estimate is \$17.248 million, with the project allocated 31% of the cost.

2. Cooper 230/138 kV Transformer
3. J1380 POI – Cooper 230 kV
4. Cooper – Deridder Mill 230 kV
5. Nelson – Penton 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is a new line from Moss Bluff – J1424 POI. The cost estimate is \$60.95 million, with the project allocated 32.7% of the cost.

4.23. J1384



Thermal analysis determined that this generator contributes to the following constraints:

1. J1217 POI – Westfork 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the J1217 POI - Westfork line. The cost estimate is \$20.448 million, with the project allocated 26.2% of the cost.

2. Westfork – Wells 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the Westfork - Wells line. The cost estimate is \$14.764 million, with the project allocated 30.9% of the cost.

4.24. J1400

Thermal analysis determined that this generator contributes to the following constraints:

1. J1400 POI – J1469 POI 161 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the J1400 POI – J1469 POI line and construct a new J1400 POI – J1469 POI line. The cost estimate is \$4 million, with the project allocated 30.1% of the cost.

2. J1007 POI – J1469 POI 161 kV
3. J1007 POI – Marked Tree 161 kV
4. Parkin – Kuhn Rd 161 kV
5. Parkin – Twist 161 kV
6. Twist – Marked Tree 161 kV
7. Powerline Rd – Trumann W 161 kV
8. Ebony – WM Lehigh-West 161 kV
9. Ebony – WM Lehigh-East 161 kV
10. W Memphis – WM Lehigh-West 161 kV
11. W Memphis – WM Polk Ave 161 kV
12. W Memphis 500/161 kV Transformer
13. Marked Tree – Harrisburg 161 kV
14. Harrisburg – Trumann 161 kV
15. Trumann – Trumann West 161 kV
16. WM Lehigh-East – WM Polk Ave 161 kV
17. Ebony – Marion 161 kV
18. J1400 POI – Marion 161 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraints. The planned upgrade is a new line from West Memphis – J1469 POI – Sans Souci 500 kV and two 500/161 kV autotransformers at the J1469 POI. The cost estimate is \$155 million, with the project allocated 28% of the cost.

4.25. J1404

Thermal analysis determined that this generator contributes to the following constraints:

1. Delta – Drew 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the Delta - Drew line. The cost estimate is \$23.7 million, with the project allocated 100% of the cost.

2. Marks – Batesville 115 kV
3. Webb – Marks 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to a new line from Ruleville - Drew. The cost estimate is \$42.8 million, with the project allocated 100% of the cost.

4. Ruleville – Schlater 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the Ruleville - Schlater line. The cost estimate is \$20.5 million, with the project allocated 100% of the cost.

4.26. J1414

Thermal analysis determined that this generator contributes to the following constraints:

1. Mt. Home – Midway Jordan 161 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the Mt. Home - Jordan line. The cost estimate is \$6.9 million, with the project allocated 100% of the cost.

4.27. J1421

Thermal analysis determined that this generator contributes to the following constraints:

1. Coly – Moler 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the Coly - Moler line. The cost estimate is \$12.5 million, with the project allocated 100% of the cost.

4.28. J1424

Thermal analysis determined that this generator contributes to the following constraints:

1. J1282 POI – Penton 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to reconductor the J1282 – Penton line. The cost estimate is \$17.248 million, with the project allocated 31% of the cost.

2. Cooper 230/138 kV Transformer
3. J1380 POI – Cooper 230 kV
4. Cooper – Deridder Mill 230 kV
5. Nelson – Penton 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is a new line from Moss Bluff – J1424 POI. The cost estimate is \$60.95 million, with the project allocated 32.7% of the cost.

4.29. J1434

Thermal analysis determined that this generator contributes to the following constraints:

1. Brinkley East – Devalls Bluff 115 kV



Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the Brinkley East – Devalls Bluff line. The cost estimate is \$35.1 million, with the project allocated 30.9% of the cost.

4.30. J1462

Thermal analysis determined that this generator contributes to the following constraints:

1. J1462 POI – Scanlan 138 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the J1462 POI - Scanlan line. The cost estimate is \$2.9 million, with the project allocated 100% of the cost.

4.31. J1469

Thermal analysis determined that this generator contributes to the following constraints:

1. J1400 POI – J1469 POI 161 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the J1400 POI – J1469 POI line and construct a new J1400 POI – J1469 POI line. The cost estimate is \$4 million, with the project allocated 39.9% of the cost.

2. J1007 POI – J1469 POI 161 kV
3. J1007 POI – Marked Tree 161 kV
4. Moses – Forest City North 161 kV
5. Forest City North – Wynne S 161 kV
6. Parkin – Kuhn Rd 161 kV
7. Parkin – Twist 161 kV
8. Parkin – Gilmore 161 kV
9. Twist – Marked Tree 161 kV
10. Powerline Rd – Trumann W 161 kV
11. Ebony – WM Lehigh-West 161 kV
12. Ebony – WM Lehigh-East 161 kV
13. W Memphis – WM Lehigh-West 161 kV
14. W Memphis – WM Polk Ave 161 kV
15. W Memphis 500/161 kV Transformer
16. Marked Tree – Harrisburg 161 kV
17. Harrisburg – Trumann 161 kV
18. Trumann – Trumann West 161 kV
19. Gilmore – Wilson 161 kV
20. WM Lehigh-East – WM Polk Ave 161 kV
21. Ebony – Marion 161 kV
22. J1400 POI – Marion 161 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraints. The planned upgrade is a new line from West Memphis – J1469 POI – Sans Souci 500 kV and two 500/161 kV autotransformers at the J1469 POI. The cost estimate is \$155 million, with the project allocated 39.1% of the cost.

5. Voltage Analysis

The voltage analysis results for 2024 Summer show no generator projects causing voltage violations. The details pertaining to the voltage analysis can be found in Appendix C – MISO ERIIS Analysis (CEII).



Although a low voltage was initially identified on the Fourchon 115 kV bus, it was determined that changing the settings of existing equipment would be sufficient to mitigate the constraint and no network upgrades were required.

6. Stability Analysis

Stability analysis was not included as part of the DPP 2019 Cycle 1 South Phase I analysis. Stability analysis will be completed as part of the Phase II and Phase III analyses.

7. Short Circuit Analysis

Short circuit analysis was not included as part of the DPP 2019 Cycle 1 South Phase I analysis. Short circuit analysis will be completed as part of the Phase II analysis.

8. Affected System Impact Study

Affected System Impact Studies were not included as part of the DPP 2019 Cycle 1 South Phase I analysis. Affected System Impact Studies will be completed as part of the Phase II and Phase III analyses.

9. Deliverability Analysis

9.1. Introduction

Generator interconnection projects have to pass Generator Deliverability Study to be granted NRIS. If the generator is deemed not fully deliverable, the customer can choose either to change the project to an Energy Resource project or to proceed with the system upgrades that will make the generator fully deliverable. Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottled up. The wind and solar generators are tested at 100% of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TEMT).

The MISO Generator Deliverability Study whitepaper describing the algorithm can be found in BPM 015 – Generation Interconnection, Appendix C.

9.2. Determining the MW Restriction

If one facility is overloaded based on the assessed “severe yet credible dispatch” scenario described in the study methodology, and the generator under study has a DF greater than 5%, part or all of its output is not deliverable. The restricted MW is calculated as following:

$$(\text{MW restricted}) = (\text{worst loading} - \text{MW rating}) / (\text{generator sensitivity factor})$$

If the result is larger than the maximum output of the generator, 100% of this generator's output is not deliverable.

9.3. Deliverability Study Results

The limiting constraints (monitored facilities – contingency facility pairs) seen in the deliverability analysis for the 2024 Summer case are summarized in Appendix D – MISO Deliverability Analysis (CEII).

9.3.1. J1205

This generator is determined to be deliverable up to 3.9 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 6 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 6: NRIS Results for J1205

J1205 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)	3.9 MW (3.10%)
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Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Cocodrie - Ville Platte 230 kV	7.8	21.6%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	2,339,072	22,600,000
J1384 POI - J1217 POI 230 kV	15.9	21.6%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1465	285,759	4,050,000
West Memphis - Birmingham Steel 500 kV	16.4	5.2%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	369,585	31,500,000
Big Cajun - Fancy Point 500 kV	24.0	30.6%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	1,412,226	29,600,000
Ville Platte - J1384 POI 230 kV	33.6	21.6%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	3,309,237	360,750,000
Manena - Sadie 230 kV	51.0	10.5%	J1205 J1282 J1380 J1424	409,979	9,600,000
Champagne - Bobcat 138 kV	57.7	8.9%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	605,446	198,750,000
J683_Tap - Wilbert 138 kV	59.4	8.8%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	604,078	198,750,000
Webre - Bayou Labutte 500 kV	60.4	31.1%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	4,305,134	89,900,000
Bobcat - Krotz Spring 138 kV	65.5	8.9%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	605,446	198,750,000
Champagne - E Opelusas 138 kV	67.6	5.8%	J1205 J1217 J1220 J1236 J1238 J1384 J1465	395,421	198,750,000
Rork - Sellers Rd 230 kV	74.2	5.2%	J1205 J1277 J1278 J1279 J1282 J1380 J1424 J1442	1,018,952	55,000,000
Livonia - J683_Tap 138 kV	76.0	8.9%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367	605,446	198,750,000

			J1380 J1384 J1424 J1442 J1455 J1462 J1465		
E Opelusas - Colton 138 kV	77.8	5.8%	J1205 J1217 J1220 J1236 J1238 J1384 J1465	395,421	198,750,000
Bayou Labutte - Willow Glen 500 kV	81.0	26.1%	J1205 J1217 J1220 J1236 J1238 J1246 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462 J1465	778,921	26,700,000
Krotz Spring - Line-642 Tap 138 kV	86.9	8.9%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	605,446	198,750,000
Line-642 Tap - Livonia 138 kV	92.5	8.9%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	605,446	198,750,000
Colton - Bloomfield 138 kV	97.2	5.8%	J1205 J1217 J1220 J1236 J1238 J1384 J1465	395,421	198,750,000
Plaisance - Champagne 138 kV	97.3	11.7%	J1205 J1217 J1220 J1236 J1238 J1286 J1290 J1384	799,052	198,750,000
Nelson 500/230 kV Transformer	100.9	14.5%	J1205 J1277 J1278 J1279 J1282 J1380 J1424 J1442 J1452	990,420	162,000,000
J1060 POI - West Memphis 500 kV	112.9	10.6%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	19,292	1,000,000
Manuel - Eunice 138 kV	114.1	7.3%	J1205 J1217 J1220 J1236 J1238 J1286 J1290 J1384	498,039	198,750,000
Nelson - Richard 500 kV	114.3	12.4%	J1205 J1277 J1278 J1279 J1380 J1282 J1424 J1442 J1452	846,683	162,000,000
Coughlin - Manuel RCT 138 kV	114.3	7.3%	J1205 J1217 J1220 J1236 J1238 J1384	498,039	198,750,000
Manuel RCT - Manuel 138 kV	114.3	7.3%	J1205 J1217 J1220 J1236 J1238 J1384	498,039	198,750,000
Richard - Wells 500 kV	118.6	25.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1462	1,721,428	162,000,000
Sadie - Rose Bluff 230 kV	123.0	10.5%	J1205 J1282 J1380 J1424	409,979	9,600,000
Bayou Vista - Terrebonne 230 kV	125.0	13.1%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	892,776	198,750,000

9.3.2. J1215

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 7 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 7: NRIS Results for J1215

J1215 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Ray Braswell - Franklin 500 kV	25.5	6.0%	J1215 J1239 J1245 J1260 J1261 J1280 J1281 J1287 J1293 J1294 J1347 J1351 J1373 J1404 J1425 J1434 J1436 J1439 J1440 J1458	173,322	3,000,000
West Memphis - Birmingham Steel 500 kV	113.2	5.3%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	603,844	31,500,000
Perryville - Baxter Wilson 500 kV	141.2	6.4%	J1215 J1239 J1245 J1258 J1260 J1261 J1280 J1281 J1373 J1415 J1425 J1436 J1437 J1452	79,682	1,000,000
Sans Souci - Driver 500 kV	161.5	6.0%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	79,701	3,300,000
J1060 POI - West Memphis 500 kV	162.7	10.2%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	27,655	1,000,000
Wabaseka - Humphrey 115 kV	165.0	19.6%	J1215 J1245 J1261 J1280 J1293 J1434	4,868,105	10,300,000
El Dorado - Mt. Olive 500 kV	165.5	8.7%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415	173,881	2,000,000

			J1425 J1434 J1236 J1437 J1469		
Horn Lake - Freeport 161 kV	167.8	5.2%	J1215 J1245 J1261 J1280 J1287 J1293 J1404 J1434 J1439 J1440	1,492,661	10,100,000
Wabbaseka - Wabbaseka West 115 kV	187.0	19.6%	J1215 J1245 J1261 J1280 J1293 J1434	1,890,738	4,000,000
Crossroads - Clarkmun 230/115 kV	191.9	6.0%	J1215 J1245 J1261 J1280 J1293 J1434	1,996,236	6,000,000
Wabbaseka West - Alzheimer 115 kV	197.1	19.6%	J1215 J1245 J1261 J1280 J1293 J1434	3,214,255	6,800,000
Alzheimer - Woodward 115 kV	200.0	19.6%	J1215 J1245 J1261 J1280 J1293 J1434	10,399,061	22,000,000

9.3.3. J1217

This generator is determined to be deliverable up to 26.2 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 8 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 8: NRIS Results for J1217

J1217 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			26.2 MW (13.09%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainabl e (MW)	Distribu tion Factor	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Big Cajun - Fancy Point 500 kV	81.7	38.1%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	2,815,762	29,600,000
Champagne - Bobcat 138 kV	92.3	9.5%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	1,037,673	198,750,000
J683_Tap - Wilbert 138 kV	95.1	9.5%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	1,036,578	198,750,000
Webre - Bayou Labutte 500 kV	96.6	38.7%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	8,568,158	89,900,000
Bobcat - Krotz Spring 138 kV	104.8	9.5%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	1,037,673	198,750,000
Champagne - E Opelusas 138 kV	118.6	12.1%	J1205 J1217 J1220 J1236 J1238 J1384 J1465	1,323,361	198,750,000
Livonia - J683_Tap 138 kV	120.4	9.5%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	1,037,673	198,750,000
East Leesville - Rodemacher 230 kV	121.6	14.2%	J1217 J1220 J1236 J1238 J1384	16,682,877	67,408,000
E Opelusas - Colton 138 kV	124.5	12.1%	J1205 J1217 J1220 J1236 J1238 J1384 J1465	1,323,361	198,750,000

Bayou Labutte - Willow Glen 500 kV	125.2	32.4%	J1205 J1217 J1220 J1236 J1238 J1246 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462 J1465	1,550,195	26,700,000
Cocodrie - Coughlin 138 kV	129.7	32.8%	J1217 J1220 J1236 J1238 J1384	3,586,976	198,750,000
Krotz Spring - Line-642 Tap 138 kV	139.0	9.5%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	1,037,673	198,750,000
Line-642 Tap - Livonia 138 kv	147.9	9.5%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	1,037,673	198,750,000
Colton - Bloomfield 138 kV	155.6	12.1%	J1205 J1217 J1220 J1236 J1238 J1384 J1465	1,323,361	198,750,000
Plaisance - Champagne 138 kV	161.5	21.6%	J1205 J1217 J1220 J1236 J1238 J1286 J1290 J1384	2,363,223	198,750,000
J1060 POI - West Memphis 500 kV	180.6	9.4%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	27,521	1,000,000
Manuel - Eunice 138 kV	182.9	14.4%	J1205 J1217 J1220 J1236 J1238 J1286 J1290 J1384	1,571,833	198,750,000
Coughlin - Manuel RCT 138 kV	182.9	14.4%	J1205 J1217 J1220 J1236 J1238 J1384	1,571,833	198,750,000
Manuel RCT - Manuel 138 kV	182.9	14.4%	J1205 J1217 J1220 J1236 J1238 J1384	1,571,833	198,750,000
Richard - Wells 500 kV	188.7	17.3%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1462	1,891,312	162,000,000
Bloomfield - Vatican 138 kV	196.8	8.1%	J1217 J1220 J1236 J1238 J1384 J1462 J1465	3,420,102	217,000,000
Bayou Vista - Terrebonne 230 kV	200.0	16.7%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	1,827,968	198,750,000

9.3.4. J1219

This generator is determined to be deliverable up to 89.8 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 9 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 9: NRIS Results for J1219

J1219 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIIS and case assumptions)			80.7 MW (80.73%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribu tion Factor	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)

J1060 POI - West Memphis 500 kV	100.0	6.42%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	9,772	1,000,000
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9.3.5. J1220

This generator is determined to be deliverable up to 6.2 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 10 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 10: NRIS Results for J1220

J1220 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)			6.2 MW (3.1%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Cocodrie - Ville Platte 230 kV	12.4	30.8%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	6,913,902	22,600,000
J1384 POI - J1217 POI 230 kV	25.5	30.8%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1465	844,655	4,050,000
West Memphis - Birmingham Steel 500 kV	26.2	5.6%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	631,137	31,500,000
Big Cajun - Fancy Point 500 kV	38.3	27.6%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	2,037,674	29,600,000
Ville Platte - J1384 POI 230 kV	81.7	30.8%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	9,781,545	360,750,000
Champagne - Bobcat 138 kV	92.3	8.7%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	950,105	198,750,000
J683_Tap - Wilbert 138 kV	95.1	8.7%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279	949,011	198,750,000

			J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465		
Webre - Bayou Labutte 500 kV	96.6	28.0%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	6,210,031	89,900,000
Bobcat - Krotz Spring 138 kV	104.8	8.7%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	950,105	198,750,000
Champagne - E Opelusas 138 kV	118.6	8.9%	J1205 J1217 J1220 J1236 J1238 J1384 J1465	969,808	198,750,000
Livonia - J683_Tap 138 kV	120.4	8.7%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	950,105	198,750,000
East Leesville - Rodemacher 230 kV	121.6	20.2%	J1217 J1220 J1236 J1238 J1384	23,812,513	67,408,000
E Opelusas - Colton 138 kV	124.5	8.9%	J1205 J1217 J1220 J1236 J1238 J1384 J1465	969,808	198,750,000
Bayou Labutte - Willow Glen 500 kV	125.2	23.6%	J1205 J1217 J1220 J1236 J1238 J1246 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462 J1465	1,125,851	26,700,000
Cocodrie - Coughlin 138 kV	129.7	14.6%	J1217 J1220 J1236 J1238 J1384	1,593,725	198,750,000
Krotz Spring - Line-642 Tap 138 kV	139.0	8.7%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	950,105	198,750,000
Line-642 Tap - Livonia 138 kv	147.2	8.7%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	950,105	198,750,000
J1220 - Rapides 230 kV	147.9	36.9%	J1220, J1238, J1452	6,922,357	10,000,000
Colton - Bloomfield 138 kV	155.6	8.9%	J1205 J1217 J1220 J1236 J1238 J1384 J1465	969,808	198,750,000
Plaisance - Champagne 138 kV	161.5	16.2%	J1205 J1217 J1220 J1236 J1238 J1286 J1290 J1384	1,773,238	198,750,000
J1060 POI - West Memphis 500 kV	180.6	10.9%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	31,751	1,000,000
Manuel - Eunice 138 kV	181.9	10.7%	J1205 J1217 J1220 J1236	1,165,740	198,750,000

			J1238 J1286 J1290 J1384		
Coughlin - Manuel RCT 138 kV	181.9	10.7%	J1205 J1217 J1220 J1236 J1238 J1384	1,165,740	198,750,000
Manuel RCT - Manuel 138 kV	181.9	10.7%	J1205 J1217 J1220 J1236 J1238 J1384	1,165,740	198,750,000
Rapides - Forest Hill 230 kV	182.9	32.0%	J1452, J1220, J1238	30,081,293	42,768,000
Richard - Wells 500 kV	188.7	17.6%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1462	1,923,072	162,000,000
Bloomfield - Vatican 138 kV	196.8	5.9%	J1217 J1220 J1236 J1238 J1384 J1462 J1465	2,523,653	217,000,000
Bayou Vista - Terrebonne 230 kV	200.0	12.1%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	1,327,739	198,750,000

9.3.6. J1223

This generator is determined to be deliverable up to 134.8 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 11 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 11: NRIS Results for J1223

J1223 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			134.8 MW (89.8%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
J1060 POI - West Memphis 500 kV	150.0	6.3%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	14,437	1,000,000

9.3.7. J1236

This generator is determined to be deliverable up to 6.6 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 12 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 12: NRIS Results for J1236

J1236 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			6.6 MW (13.1%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)

Big Cajun - Fancy Point 500 kV	20.4	38.1%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	703,940	29,600,000
Champagne - Bobcat 138 kV	23.1	9.5%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	259,418	198,750,000
J683_Tap - Wilbert 138 kV	23.8	9.5%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	259,145	198,750,000
Webre - Bayou Labutte 500 kV	24.1	38.7%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	2,142,039	89,900,000
Bobcat - Krotz Spring 138 kV	26.2	9.5%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	259,418	198,750,000
Champagne - E Opelusas 138 kV	29.7	12.1%	J1205 J1217 J1220 J1236 J1238 J1384 J1465	330,840	198,750,000
Livonia - J683_Tap 138 kV	30.1	9.5%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	259,418	198,750,000
East Leesville - Rodemacher 230 kV	30.4	14.2%	J1217 J1220 J1236 J1238 J1384	4,170,719	67,408,000
E Opelusas - Colton 138 kV	31.1	12.1%	J1205 J1217 J1220 J1236 J1238 J1384 J1465	330,840	198,750,000
Bayou Labutte - Willow Glen 500 kV	31.3	32.4%	J1205 J1217 J1220 J1236 J1238 J1246 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462 J1465	387,549	26,700,000
Cocodrie - Coughlin 138 kV	32.4	32.8%	J1217 J1220 J1236 J1238 J1384	896,744	198,750,000
Krotz Spring - Line-642 Tap 138 kV	34.8	9.5%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	259,418	198,750,000
Line-642 Tap - Livonia 138 kv	37.0	9.5%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	259,418	198,750,000
Colton - Bloomfield 138 kV	38.9	12.1%	J1205 J1217 J1220 J1236 J1238 J1384 J1465	330,840	198,750,000
Plaisance - Champagne 138 kV	40.4	21.6%	J1205 J1217 J1220 J1236 J1238 J1286 J1290 J1384	590,806	198,750,000
J1060 POI - West Memphis 500 kV	45.2	9.4%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	6,880	1,000,000
Manuel - Eunice 138 kV	45.7	14.4%	J1205 J1217 J1220 J1236 J1238 J1286 J1290 J1384	392,958	198,750,000
Coughlin - Manuel RCT 138 kV	45.7	14.4%	J1205 J1217 J1220 J1236 J1238	392,958	198,750,000

			J1384		
Manuel RCT - Manuel 138 kV	45.7	14.4%	J1205 J1217 J1220 J1236 J1238 J1384	392,958	198,750,000
Richard - Wells 500 kV	47.2	17.3%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1462	472,828	162,000,000
Bloomfield - Vatican 138 kV	49.2	8.1%	J1217 J1220 J1236 J1238 J1384 J1462 J1465	855,025	217,000,000
Bayou Vista - Terrebonne 230 kV	50.0	16.7%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	456,992	198,750,000

9.3.8. J1238

This generator is determined to be deliverable up to 1.6 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 13 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 13: NRIS Results for J1238

J1238 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			1.6 MW (3.1%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distrib ution Factor	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Cocodrie - Ville Platte 230 kV	3.1	30.8%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	1,728,475	22,600,000
J1384 POI - J1217 POI 230 kV	6.4	30.8%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1465	211,164	4,050,000
West Memphis - Birmingham Steel 500 kV	6.6	5.6%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	157,784	31,500,000
Big Cajun - Fancy Point 500 kV	9.6	27.6%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	509,418	29,600,000
Ville Platte - J1384 POI 230 kV	20.4	30.8%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	2,445,386	360,750,000
Champagne - Bobcat 138 kV	23.1	8.7%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	237,526	198,750,000
J683_Tap - Wilbert 138 kV	23.8	8.7%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	237,253	198,750,000
Webre - Bayou Labutte 500 kV	24.1	28.0%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286	1,552,508	89,900,000

			J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462		
Bobcat - Krotz Spring 138 kV	26.2	8.7%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	237,526	198,750,000
Champagne - E Opelusas 138 kV	29.7	8.9%	J1205 J1217 J1220 J1236 J1238 J1384 J1465	242,452	198,750,000
Livonia - J683_Tap 138 kV	30.1	8.7%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	237,526	198,750,000
East Leesville - Rodemacher 230 kV	30.4	20.2%	J1217 J1220 J1236 J1238 J1384	5,953,128	67,408,000
E Opelusas - Colton 138 kV	31.1	8.9%	J1205 J1217 J1220 J1236 J1238 J1384 J1465	242,452	198,750,000
Bayou Labutte - Willow Glen 500 kV	31.3	23.6%	J1205 J1217 J1220 J1236 J1238 J1246 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462 J1465	281,463	26,700,000
Cocodrie - Coughlin 138 kV	32.4	14.6%	J1217 J1220 J1236 J1238 J1384	398,431	198,750,000
Krotz Spring - Line-642 Tap 138 kV	34.8	8.7%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	237,526	198,750,000
Line-642 Tap - Livonia 138 kv	36.8	8.7%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	237,526	198,750,000
J1220 - Rapides 230 kV	37.0	36.9%	J1220, J1238, J1452	1,730,589	10,000,000
Colton - Bloomfield 138 kV	38.9	8.9%	J1205 J1217 J1220 J1236 J1238 J1384 J1465	242,452	198,750,000
Plaisance - Champagne 138 kV	40.4	16.2%	J1205 J1217 J1220 J1236 J1238 J1286 J1290 J1384	443,310	198,750,000
J1060 POI - West Memphis 500 kV	45.2	10.9%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	7,938	1,000,000
Manuel - Eunice 138 kV	45.2	10.7%	J1205 J1217 J1220 J1236 J1238 J1286 J1290 J1384	291,435	198,750,000
Coughlin - Manuel RCT 138 kV	45.2	10.7%	J1205 J1217 J1220 J1236 J1238 J1384	291,435	198,750,000
Manuel RCT - Manuel 138 kV	45.2	10.7%	J1205 J1217 J1220 J1236 J1238 J1384	291,435	198,750,000
Richard - Wells 500 kV	45.5	17.6%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1462	480,768	162,000,000
Rapides - Forest Hill 230 kV	47.2	32.0%	J1452, J1220, J1238	7,520,323	42,768,000
Bloomfield - Vatican 138 kV	49.2	5.9%	J1217 J1220 J1236 J1238 J1384 J1462 J1465	630,913	217,000,000
Bayou Vista - Terrebonne 230 kV	50.0	12.1%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424	331,935	198,750,000

			J1442 J1452 J1455 J1462		
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9.3.9. J1239

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 14 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 14: NRIS Results for J1239

J1239 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			0 MW (0.0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Ray Braswell - Franklin 500 kV	0.0	8.8%	J1215 J1239 J1245 J1260 J1261 J1280 J1281 J1287 J1293 J1294 J1347 J1351 J1373 J1404 J1425 J1434 J1436 J1439 J1440 J1458	127,113	3,000,000
Bogalusa - Adams Creek 230 kV	12.7	7.6%	J1239 J1260 J1281 J1294 J1347 J1351 J1404 J1436 J1458	2,791,470	66,500,000
West Memphis - Birmingham Steel 500 kV	33.4	7.2%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	408,249	31,500,000
Bogalusa 500/230 kV Transformer	56.6	7.6%	J1239 J1260 J1281 J1294 J1347 J1351 J1404 J1436 J1458	2,791,470	66,500,000
Perryville - Baxter Wilson 500 kV	70.6	30.5%	J1215 J1239 J1245 J1258 J1260 J1261 J1280 J1281 J1373 J1415 J1425 J1436 J1437 J1452	183,304	1,000,000
Rilla - Riverton 115 kV	70.6	8.7%	J1239	33,700,000	33,700,000
Sans Souci - Driver 500 kV	80.7	6.6%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	43,282	3,300,000
J1060 POI - West Memphis 500 kV	97.0	12.9%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	18,713	1,000,000

El Dorado - Mt. Olive 500 kV	100.0	10.0%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1434 J1236 J1437 J1469	99,647	2,000,000
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9.3.10. J1245

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 15 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 15: NRIS Results for J1245

J1245 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			0 MW (0.0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Ray Braswell - Franklin 500 kV	6.4	6.0%	J1215 J1239 J1245 J1260 J1261 J1280 J1281 J1287 J1293 J1294 J1347 J1351 J1373 J1404 J1425 J1434 J1436 J1439 J1440 J1458	43,331	3,000,000
West Memphis - Birmingham Steel 500 kV	28.3	5.3%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	150,961	31,500,000
Perryville - Baxter Wilson 500 kV	35.3	6.4%	J1215 J1239 J1245 J1258 J1260 J1261 J1280 J1281 J1373 J1415 J1425 J1436 J1437 J1452	19,920	1,000,000
Sans Souci - Driver 500 kV	40.4	6.0%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	19,925	3,300,000
J1060 POI - West Memphis 500 kV	40.7	10.2%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	6,914	1,000,000
Wabaseka - Humphrey 115 kV	41.3	19.6%	J1215 J1245 J1261 J1280 J1293 J1434	1,217,026	10,300,000
El Dorado - Mt. Olive 500 kV	41.4	8.7%	J1215 J1239 J1245 J1250	43,470	2,000,000

			J1258 J1260 J1261 J1267 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1434 J1236 J1437 J1469		
Horn Lake - Freeport 161 kV	41.9	5.2%	J1215 J1245 J1261 J1280 J1287 J1293 J1404 J1434 J1439 J1440	373,165	10,100,000
Wabbaseka - Wabbaseka West 115 kV	46.7	19.6%	J1215 J1245 J1261 J1280 J1293 J1434	472,685	4,000,000
Crossroads - Clarkmun 230/115 kV	48.0	6.0%	J1215 J1245 J1261 J1280 J1293 J1434	494,077	6,000,000
Wabbaseka West - Alzheimer 115 kV	49.3	19.6%	J1215 J1245 J1261 J1280 J1293 J1434	803,564	6,800,000
Alzheimer - Woodward 115 kV	50.0	19.6%	J1215 J1245 J1261 J1280 J1293 J1434	2,599,765	22,000,000

9.3.11. J1246

This generator is determined to be deliverable up to 451.4 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 16 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 16: NRIS Results for J1246

J1246 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIIS and case assumptions)			451.4 MW (62.3%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distrib ution Factor	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Bayou Labutte - Willow Glen 500 kV	585.4	58.9%	J1205 J1217 J1220 J1236 J1238 J1246 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462 J1465	10,212,770	26,700,000
J1060 POI - West Memphis 500 kV	725.2	7.4%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	80,183	1,000,000

9.3.12. J1249

This generator is determined to be deliverable up to 367.6 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 17 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 17: NRIS Results for J1249

J1249 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIIS and case assumptions)			367.6 MW (61.27%)		
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Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribu tion Factor	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Purvis Bulk 230/161 kV Transformer #1	508.6	50.6%	J1249	0	0
Purvis Bulk 230/161 kV Transformer #2	600.0	50.4%	J1249	0	0

9.3.13. J1250

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 18 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 18: NRIS Results for J1250

J1250 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribu tion Factor	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
J1007 POI - J1469 POI 161 kV	4.5	6.3%	J1250 J1267 J1414	113,166	1,900,000
West Memphis - Birmingham Steel 500 kV	24.7	18.5%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	368,163	31,500,000
Sans Souci - Driver 500 kV	28.9	22.6%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	52,096	3,300,000
El Dorado - Mt. Olive 500 kV	35.0	6.4%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1434 J1236 J1437 J1469	22,270	2,000,000

9.3.14. J1257

This generator is determined to be deliverable up to 18 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 19 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 19: NRIS Results for J1257

J1257 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)			16.2 MW (80.75%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
J1060 POI - West Memphis 500 kV	20.0	6.4%	J1205 J1215 J1217 J1219 J1220	1,954	1,000,000

			J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465		
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9.3.15. J1258

This generator is determined to be deliverable up to 6.4 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 20 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 20: NRIS Results for J1258

J1258 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			6.4 MW (12.7%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
West Memphis - Birmingham Steel 500 kV	28.3	22.4%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	635,970	31,500,000
Perryville - Baxter Wilson 500 kV	35.3	10.1%	J1215 J1239 J1245 J1258 J1260 J1261 J1280 J1281 J1373 J1415 J1425 J1436 J1437 J1452	31,027	1,000,000
Sans Souci - Driver 500 kV	40.4	21.4%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	70,530	3,300,000
J1060 POI - West Memphis 500 kV	41.3	28.1%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	20,714	1,000,000
El Dorado - Mt. Olive 500 kV	50.0	12.4%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415	62,029	2,000,000

			J1425 J1434 J1236 J1437 J1469		
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9.3.16. J1259

This generator is determined to be deliverable up to 44.9 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 21 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 21: NRIS Results for J1259

J1259 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			40.4 MW (80.72%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
J1060 POI - West Memphis 500 kV	50.0	6.3%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	4,812	1,000,000

9.3.17. J1260

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 22 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 22: NRIS Results for J1260

J1260 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Ray Braswell - Franklin 500 kV	0.0	9.4%	J1215 J1239 J1245 J1260 J1261 J1280 J1281 J1287 J1293 J1294 J1347 J1351 J1373 J1404 J1425 J1434 J1436 J1439 J1440 J1458	221,686	3,000,000
Bogalusa - Adams Creek 230 kV	20.9	6.1%	J1239 J1260 J1281 J1294 J1347 J1351 J1404 J1436 J1458	3,670,820	66,500,000
West Memphis - Birmingham Steel 500 kV	54.8	5.3%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	495,152	31,500,000
Bogalusa 500/230 kV Transformer	92.8	6.1%	J1239 J1260 J1281 J1294	3,670,820	66,500,000

			J1347 J1351 J1404 J1436 J1458		
Perryville - Baxter Wilson 500 kV	93.2	5.5%	J1215 J1239 J1245 J1258 J1260 J1261 J1280 J1281 J1373 J1415 J1425 J1436 J1437 J1452	57,233	1,000,000
Andrus - Refuge 115 kV	108.6	6.2%	J1260 J1458	0	0
Andrus 230/115 kV Transformer	115.8	12.4%	J1260 J1281 J1458	4,844,635	18,000,000
Sans Souci - Driver 500 kV	123.4	5.3%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	56,807	3,300,000
J620_Tap - Macon Lake 115 kV	127.8	31.0%	J1260 J1281 J1425	6,325,020	7,900,000
Refuge - Greenville Industrial 115 kV	132.4	6.2%	J1260 J1458	0	0
J1060 POI - West Memphis 500 kV	159.0	10.9%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	25,114	1,000,000
El Dorado - Mt. Olive 500 kV	164.0	6.4%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1434 J1236 J1437 J1469	104,517	2,000,000

9.3.18. J1261

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 23 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 23: NRIS Results for J1261

J1261 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Ray Braswell - Franklin 500 kV	11.5	6.0%	J1215 J1239 J1245 J1260 J1261 J1280 J1281 J1287 J1293 J1294 J1347 J1351 J1373 J1404 J1425 J1434 J1436 J1439 J1440 J1458	77,088	3,000,000
West Memphis - Birmingham Steel 500 kV	51.0	8.3%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281	426,273	31,500,000

			J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469		
Perryville - Baxter Wilson 500 kV	63.5	5.0%	J1215 J1239 J1245 J1258 J1260 J1261 J1280 J1281 J1373 J1415 J1425 J1436 J1437 J1452	26,799	1,000,000
Sans Souci - Driver 500 kV	73.2	9.3%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	54,986	3,300,000
Wabbaseka - Humphrey 115 kV	74.3	11.2%	J1215 J1245 J1261 J1280 J1293 J1434	1,244,017	10,300,000
El Dorado - Mt. Olive 500 kV	74.5	7.5%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1434 J1236 J1437 J1469	67,532	2,000,000
Horn Lake - Freeport 161 kV	75.5	5.2%	J1215 J1245 J1261 J1280 J1287 J1293 J1404 J1434 J1439 J1440	670,408	10,100,000
Wabbaseka - Wabbaseka West 115 kV	77.6	11.2%	J1215 J1245 J1261 J1280 J1293 J1434	483,280	4,000,000
Wynne South - Wynne East 161 kV	84.1	13.7%	J1261, J1280, J1293, J1373, J1434	1,658,530	6,100,000
Crossroads - Clarkmun 230/115 kV	86.4	6.1%	J1215 J1245 J1261 J1280 J1293 J1434	907,274	6,000,000
Wabbaseka West - Alzheimer 115 kV	88.7	11.2%	J1215 J1245 J1261 J1280 J1293 J1434	821,575	6,800,000
Alzheimer - Woodward 115 kV	90.0	11.2%	J1215 J1245 J1261 J1280 J1293 J1434	2,658,038	22,000,000

9.3.19. J1267

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 24 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 24: NRIS Results for J1267

J1267 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
J1007 POI - J1469 POI 161 kV	17.2	12.7%	J1250 J1267 J1414	1,301,836	1,900,000
West Memphis - Birmingham Steel 500 kV	95.3	30.1%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	2,306,634	31,500,000

Sans Souci - Driver 500 kV	98.9	31.1%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	277,275	3,300,000
Kuhn Rd - Ebony 161 kV	111.4	21.5%	J1267	4,000,000	4,000,000
El Dorado - Mt. Olive 500 kV	135.0	7.4%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1434 J1236 J1437 J1469	100,082	2,000,000

9.3.20. J1277

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 25 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 25: NRIS Results for J1277

J1277 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Nelson - Goosport 138 kV #2	0.0	5.2%	J1277 J1279 J1282 J1380 J1424	368,735	162,000,000
Nelson - LC Bulk South 138 kV	0.0	5.2%	J1277 J1279 J1282 J1380 J1424	370,158	162,000,000
Goosport - LC Bulk North 138 kV	0.0	5.2%	J1277 J1279 J1282 J1380 J1424	368,735	162,000,000
LC Bulk North - Henning 138 kV	0.0	6.8%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	483,341	162,000,000
LC Bulk South - Lowegrout 138 kV	4.0	6.2%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	442,055	162,000,000
Cocodrie - Ville Platte 230 kV	8.1	7.4%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	745,485	22,600,000
J1384 POI - J1217 POI 230 kV	16.6	7.4%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1465	91,074	4,050,000
West Memphis - Birmingham Steel 500 kV	17.0	6.2%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	460,502	31,500,000
Big Cajun - Fancy Point 500 kV	24.9	28.3%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	1,360,545	29,600,000
Ville Platte - J1384 POI 230 kV	27.3	7.4%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	1,054,686	360,750,000
Lowegrout - Jennings 138 kV	53.1	6.2%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	442,055	162,000,000
Champagne - Bobcat 138 kV	53.1	6.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	442,543	198,750,000

Bayou Cove - Richard 138 kV #1	60.0	6.1%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	434,936	162,000,000
J683_Tap - Wilbert 138 kV	61.8	6.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	441,832	198,750,000
Webre - Bayou Labutte 500 kV	62.8	28.9%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	4,163,291	89,900,000
Bobcat - Krotz Spring 138 kV	69.5	6.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	442,543	198,750,000
Sabine - Mud Lake 230 kV	70.3	18.1%	J1277 J1278 J1279 J1442	113,653	400,000
Rork - Sellers Rd 230 kV	72.3	12.9%	J1205 J1277 J1278 J1279 J1282 J1380 J1424 J1442	2,652,350	55,000,000
Henning - Hebert 138 kV	77.1	6.8%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	483,341	162,000,000
Livonia - J683_Tap 138 kV	80.4	6.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	442,543	198,750,000
J1282 POI - J1424 POI 230 kV	80.9	9.3%	J1277, J1278, J1279, J1282, J1442	331,428	4,800,000
Bayou Labutte - Willow Glen 500 kV	84.3	24.0%	J1205 J1217 J1220 J1236 J1238 J1246 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462 J1465	746,713	26,700,000
Krotz Spring - Line-642 Tap 138 kV	84.6	6.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	442,543	198,750,000
Cooper 230/138 kV Transformer	87.9	7.4%	J1277 J1279 J1380 J1242 J1424 J1442	527,476	162,000,000
Hebert - Bayou Cove 138 kV	90.4	6.8%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	483,341	162,000,000
Line-642 Tap - Livonia 138 kv	91.8	6.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	442,543	198,750,000
Sans Souci - Driver 500 kV	101.2	5.7%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	48,461	3,300,000
Nelson 500/230 kV Transformer	104.9	27.4%	J1205 J1277 J1278 J1279 J1282 J1380 J1424 J1442 J1452	1,951,162	162,000,000
J1060 POI - West Memphis 500 kV	109.3	11.4%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	21,352	1,000,000
J1380 POI - Cooper 230 kV	116.1	7.4%	J1277 J1279 J1282 J1380 J1424 J1442	527,476	162,000,000

Bayou Cove - Richard 138 kV #2	118.7	6.9%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	489,748	162,000,000
Nelson - Richard 500 kV	118.9	34.5%	J1205 J1277 J1278 J1279 J1380 J1282 J1424 J1442 J1452	2,455,147	162,000,000
Richard - Wells 500 kV	127.9	31.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1462	2,341,964	162,000,000
Bayou Vista - Terrebonne 230 kV	130.0	11.4%	J1205 J1217 J1220 J1236 J1238 J1277 J1282 J1380 J1384 J1424	813,938	198,750,000

9.3.21. J1278

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 26 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 26: NRIS Results for J1278

J1278 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
LC Bulk North - Henning 138 kV	0.0	6.0%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	327,995	162,000,000
LC Bulk South - Lowegrout 138 kV	3.1	5.5%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	300,069	162,000,000
Cocodrie - Ville Platte 230 kV	6.2	8.0%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	619,109	22,600,000
J1384 POI - J1217 POI 230 kV	12.7	8.0%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1465	75,635	4,050,000
West Memphis - Birmingham Steel 500 kV	13.1	6.6%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	373,565	31,500,000
Big Cajun - Fancy Point 500 kV	19.2	26.4%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	977,048	29,600,000
Ville Platte - J1384 POI 230 kV	21.0	8.0%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	875,894	360,750,000
Lowegrout - Jennings 138 kV	40.8	5.5%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	300,069	162,000,000
Champagne - Bobcat 138 kV	40.9	5.9%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	325,094	198,750,000
Bayou Cove - Richard 138 kV #1	46.2	5.4%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	295,141	162,000,000

J683_Tap - Wilbert 138 kV	47.6	5.9%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	323,999	198,750,000
Webre - Bayou Labutte 500 kV	48.3	27.0%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	2,989,768	89,900,000
Bobcat - Krotz Spring 138 kV	53.4	5.9%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	325,094	198,750,000
Sabine - Mud Lake 230 kV	54.1	13.9%	J1277 J1278 J1279 J1442	66,946	400,000
Rork - Sellers Rd 230 kV	55.6	11.5%	J1205 J1277 J1278 J1279 J1282 J1380 J1424 J1442	1,811,646	55,000,000
Henning - Hebert 138 kV	59.3	6.0%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	327,995	162,000,000
Livonia - J683_Tap 138 kV	61.8	5.9%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	325,094	198,750,000
J1282 POI - J1424 POI 230 kV	62.2	6.8%	J1277, J1278, J1279, J1282, J1442	186,632	4,800,000
Bayou Labutte - Willow Glen 500 kV	64.8	22.4%	J1205 J1217 J1220 J1236 J1238 J1246 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462 J1465	535,926	26,700,000
Krotz Spring - Line-642 Tap 138 kV	67.6	5.9%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	325,094	198,750,000
Hebert - Bayou Cove 138 kV	69.5	6.0%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	327,995	162,000,000
Line-642 Tap - Livonia 138 kv	70.6	5.9%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	325,094	198,750,000
Sans Souci - Driver 500 kV	72.2	6.0%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	39,653	3,300,000
J1278 POI - Moscow 138 kV	76.7	46.5%	J1278 J1442	5,515,004	7,000,000
Pelican Road - Jacinto 138 kV	77.9	9.5%	J1278 J1442	1,003,535	13,700,000
Nelson 500/230 kV Transformer	80.7	24.5%	J1205 J1277 J1278 J1279 J1282 J1380 J1424 J1442 J1452	1,340,455	162,000,000
J1060 POI - West Memphis 500 kV	89.3	11.5%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	16,625	1,000,000
Bayou Cove - Richard 138 kV #2	91.3	6.1%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	332,376	162,000,000
Nelson - Richard 500 kV	91.5	30.6%	J1205 J1277 J1278 J1279 J1380 J1282 J1424 J1442 J1452	1,677,759	162,000,000

Richard - Wells 500 kV	93.4	28.6%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1462	1,612,598	162,000,000
Moscow - Corrigan 138 kV	98.4	46.5%	J1278 J1442	8,036,149	10,200,000
Bayou Vista - Terrebonne 230 kV	100.0	10.74%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	587,796	198,750,000

9.3.22. J1279

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 27 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 27: NRIS Results for J1279

J1279 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Nelson - Goosport 138 kV #2	0.0	5.1%	J1277 J1279 J1282 J1380 J1424	278,714	162,000,000
Nelson - LC Bulk South 138 kV	0.0	5.1%	J1277 J1279 J1282 J1380 J1424	279,262	162,000,000
Goosport - LC Bulk North 138 kV	0.0	5.1%	J1277 J1279 J1282 J1380 J1424	278,714	162,000,000
LC Bulk North - Henning 138 kV	0.0	6.7%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	367,968	162,000,000
LC Bulk South - Lowegrout 138 kV	3.1	6.2%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	336,757	162,000,000
Cocodrie - Ville Platte 230 kV	6.2	7.5%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	577,319	22,600,000
J1384 POI - J1217 POI 230 kV	12.7	7.5%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1465	70,530	4,050,000
West Memphis - Birmingham Steel 500 kV	13.1	6.2%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	353,664	31,500,000
Big Cajun - Fancy Point 500 kV	19.2	28.4%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	1,050,641	29,600,000
Ville Platte - J1384 POI 230 kV	21.0	7.5%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	816,771	360,750,000
Lowegrout - Jennings 138 kV	40.8	6.2%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	336,757	162,000,000
Champagne - Bobcat 138 kV	40.9	6.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	341,513	198,750,000

Bayou Cove - Richard 138 kV #1	46.2	6.1%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	331,828	162,000,000
J683_Tap - Wilbert 138 kV	47.6	6.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	340,418	198,750,000
Webre - Bayou Labutte 500 kV	48.3	29.0%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	3,214,721	89,900,000
Bobcat - Krotz Spring 138 kV	53.4	6.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	341,513	198,750,000
Sabine - Mud Lake 230 kV	54.1	17.7%	J1277 J1278 J1279 J1442	85,590	400,000
Rork - Sellers Rd 230 kV	55.6	13.0%	J1205 J1277 J1278 J1279 J1282 J1380 J1424 J1442	2,056,036	55,000,000
Henning - Hebert 138 kV	59.3	6.7%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	367,968	162,000,000
Livonia - J683_Tap 138 kV	61.8	6.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	341,513	198,750,000
J1282 POI - J1424 POI 230 kV	62.2	9.4%	J1277, J1278, J1279, J1282, J1442	255,491	4,800,000
Bayou Labutte - Willow Glen 500 kV	64.8	24.1%	J1205 J1217 J1220 J1236 J1238 J1246 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462 J1465	576,545	26,700,000
Krotz Spring - Line-642 Tap 138 kV	65.1	6.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	341,513	198,750,000
Cooper 230/138 kV Transformer	67.6	7.4%	J1277 J1279 J1380 J1242 J1424 J1442	403,013	162,000,000
Hebert - Bayou Cove 138 kV	69.5	6.7%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	367,968	162,000,000
Line-642 Tap - Livonia 138 kv	70.6	6.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	341,513	198,750,000
Sans Souci - Driver 500 kV	77.9	5.6%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	37,212	3,300,000
Nelson 500/230 kV Transformer	80.7	27.2%	J1205 J1277 J1278 J1279 J1282 J1380 J1424 J1442 J1452	1,487,752	162,000,000
J1060 POI - West Memphis 500 kV	84.1	11.4%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	16,438	1,000,000
J1380 POI - Cooper 230 kV	89.3	7.4%	J1277 J1279 J1282 J1380 J1424 J1442	403,013	162,000,000

Bayou Cove - Richard 138 kV #2	91.3	6.8%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	373,444	162,000,000
Nelson - Richard 500 kV	91.5	34.9%	J1205 J1277 J1278 J1279 J1380 J1282 J1424 J1442 J1452	1,908,835	162,000,000
Richard - Wells 500 kV	98.4	31.4%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1462	1,813,010	162,000,000
Bayou Vista - Terrebonne 230 kV	100.0	11.5%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	628,295	198,750,000

9.3.23. J1280

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 28 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 28: NRIS Results for J1280

J1280 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distrib ution Factor	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Ray Braswell - Franklin 500 kV	12.7	6.0%	J1215 J1239 J1245 J1260 J1261 J1280 J1281 J1287 J1293 J1294 J1347 J1351 J1373 J1404 J1425 J1434 J1436 J1439 J1440 J1458	86,517	3,000,000
West Memphis - Birmingham Steel 500 kV	56.6	7.6%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	434,404	31,500,000
Perryville - Baxter Wilson 500 kV	70.6	5.1%	J1215 J1239 J1245 J1258 J1260 J1261 J1280 J1281 J1373 J1415 J1425 J1436 J1437 J1452	32,516	1,000,000
Sans Souci - Driver 500 kV	80.7	8.5%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	56,081	3,300,000
J1060 POI - West Memphis 500 kV	81.3	5.4%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	7,483	1,000,000

Wabbaseka - Humphrey 115 kV	82.5	15.0%	J1215 J1245 J1261 J1280 J1293 J1434	1,862,758	10,300,000
El Dorado - Mt. Olive 500 kV	82.8	7.8%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1434 J1236 J1437 J1469	77,536	2,000,000
Horn Lake - Freeport 161 kV	83.9	5.2%	J1215 J1245 J1261 J1280 J1287 J1293 J1404 J1434 J1439 J1440	737,735	10,100,000
Wabbaseka - Wabbaseka West 115 kV	86.3	15.0%	J1215 J1245 J1261 J1280 J1293 J1434	723,354	4,000,000
Wynne South - Wynne East 161 kV	93.5	11.3%	J1261, J1280, J1293, J1373, J1434	1,518,164	6,100,000
Crossroads - Clarkmun 230/115 kV	96.0	6.0%	J1215 J1245 J1261 J1280 J1293 J1434	988,153	6,000,000
Wabbaseka West - Alzheimer 115 kV	98.5	15.0%	J1215 J1245 J1261 J1280 J1293 J1434	1,229,702	6,800,000
Alzheimer - Woodward 115 kV	100.0	15.0%	J1215 J1245 J1261 J1280 J1293 J1434	3,978,449	22,000,000

9.3.24. J1281

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 29 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 29: NRIS Results for J1281

J1281 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERI and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Ray Braswell - Franklin 500 kV	0.0	8.8%	J1215 J1239 J1245 J1260 J1261 J1280 J1281 J1287 J1293 J1294 J1347 J1351 J1373 J1404 J1425 J1434 J1436 J1439 J1440 J1458	95,119	3,000,000
Bogalusa - Adams Creek 230 kV	9.6	6.3%	J1239 J1260 J1281 J1294 J1347 J1351 J1404 J1436 J1458	1,719,941	66,500,000
West Memphis - Birmingham Steel 500 kV	25.1	7.4%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	314,715	31,500,000
Bogalusa 500/230 kV Transformer	42.5	6.3%	J1239 J1260 J1281 J1294 J1347 J1351 J1404 J1436 J1458	1,719,941	66,500,000
Perryville - Baxter Wilson 500 kV	49.7	14.3%	J1215 J1239 J1245 J1258 J1260 J1261 J1280 J1281 J1373 J1415 J1425 J1436 J1437 J1452	65,121	1,000,000
Andrus 230/115 kV Transformer	53.0	8.3%	J1260 J1281 J1458	1,472,270	18,000,000
Sans Souci - Driver 500 kV	56.5	7.0%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	34,787	3,300,000

J620_Tap - Macon Lake 115 kV	60.5	8.5%	J1260 J1281 J1425	787,434	7,900,000
J1060 POI - West Memphis 500 kV	72.7	13.1%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	13,965	1,000,000
El Dorado - Mt. Olive 500 kV	75.0	8.4%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1434 J1236 J1437 J1469	63,255	2,000,000

9.3.25. J1282

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 30 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 30: NRIS Results for J1282

J1282 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Nelson - Goosport 138 kV #2	0.0	5.8%	J1277 J1279 J1282 J1380 J1424	631,898	162,000,000
Nelson - LC Bulk South 138 kV	0.0	5.8%	J1277 J1279 J1282 J1380 J1424	634,088	162,000,000
Goosport - LC Bulk North 138 kV	0.0	5.8%	J1277 J1279 J1282 J1380 J1424	631,898	162,000,000
LC Bulk North - Henning 138 kV	0.0	7.7%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	838,880	162,000,000
LC Bulk South - Lowegrout 138 kV	6.2	7.0%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	767,695	162,000,000
Cocodrie - Ville Platte 230 kV	12.5	9.4%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	1,461,098	22,600,000
J1384 POI - J1217 POI 230 kV	25.5	9.4%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1465	178,499	4,050,000
West Memphis - Birmingham Steel 500 kV	26.2	5.4%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	615,216	31,500,000
Big Cajun - Fancy Point 500 kV	38.3	32.0%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	2,364,589	29,600,000
Ville Platte - J1384 POI 230 kV	42.0	9.4%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	2,067,110	360,750,000

Lowegrout - Jennings 138 kV	53.7	7.0%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	767,695	162,000,000
Manena - Sadie 230 kV	81.7	25.5%	J1205 J1282 J1380 J1424	1,595,118	9,600,000
Champagne - Bobcat 138 kV	81.7	7.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	792,484	198,750,000
Bayou Cove - Richard 138 kV #1	92.3	6.9%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	755,649	162,000,000
J683_Tap - Wilbert 138 kV	95.1	7.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	790,295	198,750,000
Webre - Bayou Labutte 500 kV	96.6	32.6%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	7,225,089	89,900,000
Bobcat - Krotz Spring 138 kV	108.2	7.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	792,484	198,750,000
Rork - Sellers Rd 230 kV	111.3	13.6%	J1205 J1277 J1278 J1279 J1282 J1380 J1424 J1442	4,288,665	55,000,000
Henning - Hebert 138 kV	118.6	7.7%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	838,880	162,000,000
Livonia - J683_Tap 138 kV	119.9	7.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	792,484	198,750,000
Scott - Bonin 138 kV	123.7	5.5%	J1282 J1286 J1290 J1424 J1462 J1465	600,139	162,000,000
J1282 POI - J1424 POI 230 kV	124.5	66.5%	J1277, J1278, J1279, J1282, J1442	3,636,449	4,800,000
Bayou Labutte - Willow Glen 500 kV	129.7	27.2%	J1205 J1217 J1220 J1236 J1238 J1246 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462 J1465	1,298,838	26,700,000
Krotz Spring - Line-642 Tap 138 kV	130.2	7.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	792,484	198,750,000
Cooper 230/138 kV Transformer	135.2	25.0%	J1277 J1279 J1380 J1242 J1424 J1442	2,733,478	162,000,000
Hebert - Bayou Cove 138 kV	139.0	7.7%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	838,880	162,000,000
Line-642 Tap - Livonia 138 kv	155.7	7.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	792,484	198,750,000
Nelson 500/230 kV Transformer	161.5	38.2%	J1205 J1277 J1278 J1279 J1282 J1380 J1424 J1442 J1452	4,181,257	162,000,000
J1060 POI - West Memphis 500 kV	168.2	10.8%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463	31,403	1,000,000

			J1465		
J1380 POI - Cooper 230 kV	178.6	25.0%	J1277 J1279 J1282 J1380 J1424 J1442	2,733,478	162,000,000
Bayou Cove - Richard 138 kV #2	182.6	7.8%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	850,926	162,000,000
Nelson - Richard 500 kV	182.9	35.5%	J1205 J1277 J1278 J1279 J1380 J1282 J1424 J1442 J1452	3,892,139	162,000,000
Richard - Wells 500 kV	189.7	34.3%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1462	3,847,238	162,000,000
Sadie - Rose Bluff 230 kV	196.8	25.5%	J1205 J1282 J1380 J1424	1,595,118	9,600,000
Bayou Vista - Terrebonne 230 kV	200.0	12.98%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	1,420,780	198,750,000

9.3.26. J1286

This generator is determined to be deliverable up to 4.7 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 31 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 31: NRIS Results for J1286

J1286 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			4.7 MW (3.1%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Cocodrie - Ville Platte 230 kV	9.3	5.0%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	1,600,784	22,600,000
J1384 POI - J1217 POI 230 kV	19.6	5.0%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1465	195,564	4,050,000
Big Cajun - Fancy Point 500 kV	28.7	35.6%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	1,973,141	29,600,000
Ville Platte - J1384 POI 230 kV	61.2	5.0%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	2,264,733	360,750,000
Champagne - Bobcat 138 kV	69.2	10.1%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	830,795	198,750,000
J683_Tap - Wilbert 138 kV	71.3	10.1%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	829,153	198,750,000
Webre - Bayou Labutte 500 kV	72.4	36.1%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	5,995,605	89,900,000
Bobcat - Krotz Spring 138 kV	74.1	10.1%	J1205 J1217 J1220 J1236 J1238	830,795	198,750,000

			J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462		
Scanlan - Bosco RCT 138 kV	89.0	6.6%	J1286 J1290 J1462	885,364	9,900,000
Livonia - J683_Tap 138 kV	89.9	10.1%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	830,795	198,750,000
Scott - Bonin 138 kV	93.4	11.1%	J1282 J1286 J1290 J1424 J1462 J1465	914,171	162,000,000
Bayou Labutte - Willow Glen 500 kV	97.2	30.4%	J1205 J1217 J1220 J1236 J1238 J1246 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462 J1465	1,088,816	26,700,000
Krotz Spring - Line-642 Tap 138 kV	104.3	10.1%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	830,795	198,750,000
Line-642 Tap - Livonia 138 kv	116.7	10.1%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	830,795	198,750,000
Plaisance - Champagne 138 kV	121.1	5.2%	J1205 J1217 J1220 J1236 J1238 J1286 J1290 J1384	426,070	198,750,000
J1060 POI - West Memphis 500 kV	122.6	9.6%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	21,042	1,000,000
J1462 POI - Scanlan 138 kV	135.5	6.6%	J1286 J1290 J1462	1,558,583	55,000,000
Manuel - Eunice 138 kV	137.2	69.6%	J1205 J1217 J1220 J1236 J1238 J1286 J1290 J1384	5,709,663	198,750,000
Richard - Wells 500 kV	147.6	25.7%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1462	2,110,889	162,000,000
Bayou Vista - Terrebonne 230 kV	150.0	16.3%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	1,334,033	198,750,000

9.3.27. J1287

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 32 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 32: NRIS Results for J1287

J1287 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)	0 MW (0%)
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Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Ray Braswell - Franklin 500 kV	41.4	6.9%	J1215 J1239 J1245 J1260 J1261 J1280 J1281 J1287 J1293 J1294 J1347 J1351 J1373 J1404 J1425 J1434 J1436 J1439 J1440 J1458	49,737	3,000,000
Horn Lake - Freeport 161 kV	50.0	13.2%	J1215 J1245 J1261 J1280 J1287 J1293 J1404 J1434 J1439 J1440	945,447	10,100,000

9.3.28. J1290

This generator is determined to be deliverable up to 0.8 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 33 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 33: NRIS Results for J1290

J1290 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			0.8 MW (3.12%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Cocodrie - Ville Platte 230 kV	1.6	5.0%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	266,797	22,600,000
J1384 POI - J1217 POI 230 kV	3.3	5.0%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1465	32,594	4,050,000
Big Cajun - Fancy Point 500 kV	4.8	35.6%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	328,857	29,600,000
Ville Platte - J1384 POI 230 kV	10.2	5.0%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	377,456	360,750,000
Champagne - Bobcat 138 kV	11.5	10.1%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	138,466	198,750,000
J683_Tap - Wilbert 138 kV	11.9	10.1%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	138,192	198,750,000
Webre - Bayou Labutte 500 kV	12.1	36.1%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	999,268	89,900,000
Bobcat - Krotz Spring 138 kV	12.4	10.1%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	138,466	198,750,000
Scanlan - Bosco RCT 138 kV	14.8	6.6%	J1286 J1290 J1462	147,561	9,900,000
Livonia - J683_Tap 138 kV	15.0	10.1%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286	138,466	198,750,000

			J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465		
Scott - Bonin 138 kV	15.6	11.1%	J1282 J1286 J1290 J1424 J1462 J1465	152,362	162,000,000
Bayou Labutte - Willow Glen 500 kV	16.2	30.4%	J1205 J1217 J1220 J1236 J1238 J1246 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462 J1465	181,469	26,700,000
Krotz Spring - Line-642 Tap 138 kV	17.4	10.1%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	138,466	198,750,000
Line-642 Tap - Livonia 138 kV	19.5	10.1%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	138,466	198,750,000
Plaisance - Champagne 138 kV	20.2	5.2%	J1205 J1217 J1220 J1236 J1238 J1286 J1290 J1384	71,012	198,750,000
J1060 POI - West Memphis 500 kV	20.4	9.6%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	3,507	1,000,000
J1462 POI - Scanlan 138 kV	22.6	6.6%	J1286 J1290 J1462	259,764	55,000,000
Manuel - Eunice 138 kV	23.4	69.6%	J1205 J1217 J1220 J1236 J1238 J1286 J1290 J1384	951,610	198,750,000
Richard - Wells 500 kV	24.6	25.7%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1462	351,815	162,000,000
Bayou Vista - Terrebonne 230 kV	25.0	16.3%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	222,339	198,750,000

9.3.29. J1293

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 34 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 34: NRIS Results for J1293

J1293 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distrib ution Factor	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Ray Braswell - Franklin 500 kV	0.0	6.0%	J1215 J1239 J1245 J1260 J1261 J1280 J1281 J1287 J1293 J1294	21,557	3,000,000

			J1347 J1351 J1373 J1404 J1425 J1434 J1436 J1439 J1440 J1458		
J552_Tap - Brinkley East 115 kV	3.2	100.0%	J1293	1,640,000	8,200,000
West Memphis - Birmingham Steel 500 kV	10.2	7.9%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	112,439	31,500,000
J552 Tap - Moro 115 kV	10.2	100.0%	J1293	1,640,000	8,200,000
Moro - Marianna West 115 kV	14.6	100.0%	J1293	1,640,000	8,200,000
Marianna West - Marianna 115 kV	17.7	100.0%	J1293	1,640,000	8,200,000
Sans Souci - Driver 500 kV	20.3	9.0%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	14,796	3,300,000
Wabaseka - Humphrey 115 kV	20.6	7.2%	J1215 J1245 J1261 J1280 J1293 J1434	222,103	10,300,000
El Dorado - Mt. Olive 500 kV	20.7	7.1%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1434 J1236 J1437 J1469	17,733	2,000,000
Horn Lake - Freeport 161 kV	21.0	5.4%	J1215 J1245 J1261 J1280 J1287 J1293 J1404 J1434 J1439 J1440	194,461	10,100,000
Wabaseka - Wabaseka West 115 kV	21.6	7.2%	J1215 J1245 J1261 J1280 J1293 J1434	86,085	4,000,000
Wynne South - Wynne East 161 kV	23.0	15.4%	J1261, J1280, J1293, J1373, J1434	531,761	6,100,000
Marianna - Helena West 115 kV	23.4	100.0%	J1293	1,640,000	8,200,000
Crossroads - Clarkmun 230/115 kV	24.0	6.5%	J1215 J1245 J1261 J1280 J1293 J1434	269,043	6,000,000
Wabaseka West - Alzheimer 115 kV	24.6	7.2%	J1215 J1245 J1261 J1280 J1293 J1434	146,344	6,800,000
Alzheimer - Woodward 115 kV	25.0	7.2%	J1215 J1245 J1261 J1280 J1293 J1434	473,467	22,000,000

9.3.30. J1294

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 35 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 35: NRIS Results for J1294

J1294 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Ray Braswell - Franklin 500 kV	0.0	13.8%	J1215 J1239 J1245 J1260 J1261 J1280 J1281 J1287 J1293 J1294 J1347 J1351 J1373 J1404 J1425 J1434 J1436 J1439 J1440 J1458	196,672	3,000,000
Bogalusa - Adams Creek 230 kV	33.1	7.8%	J1239 J1260 J1281 J1294 J1347	2,843,343	66,500,000

			J1351 J1404 J1436 J1458		
Bogalusa 500/230 kV Transformer	75.5	7.8%	J1239 J1260 J1281 J1294 J1347 J1351 J1404 J1436 J1458	2,843,343	66,500,000
Isola - Belzoni 115 kV	82.0	9.2%	J1294 J1458	230,569	500,000
J1440 POI - Tunica 115 kV	99.0	7.5%	J1294 J1404 J1439 J1440	1,161,298	28,600,000

9.3.31. J1347

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 36 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 36: NRIS Results for J1347

J1347 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERI and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Ray Braswell - Franklin 500 kV	0.0	15.4%	J1215 J1239 J1245 J1260 J1261 J1280 J1281 J1287 J1293 J1294 J1347 J1351 J1373 J1404 J1425 J1434 J1436 J1439 J1440 J1458	388,967	3,000,000
Bogalusa - Adams Creek 230 kV	58.4	7.7%	J1239 J1260 J1281 J1294 J1347 J1351 J1404 J1436 J1458	4,917,127	66,500,000
Bogalusa 500/230 kV Transformer	175.0	7.7%	J1239 J1260 J1281 J1294 J1347 J1351 J1404 J1436 J1458	4,917,127	66,500,000

9.3.32. J1351

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 37 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 37: NRIS Results for J1351

J1351 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERI and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Ray Braswell - Franklin 500 kV	0.0	13.3%	J1215 J1239 J1245 J1260 J1261 J1280 J1281 J1287 J1293 J1294 J1347 J1351 J1373 J1404 J1425 J1434 J1436 J1439 J1440 J1458	230,098	3,000,000
Bogalusa - Adams Creek 230 kV	40.1	10.8%	J1239 J1260 J1281 J1294 J1347 J1351 J1404 J1436 J1458	4,752,093	66,500,000
Bogalusa 500/230 kV Transformer	96.9	10.8%	J1239 J1260 J1281 J1294 J1347 J1351 J1404 J1436 J1458	4,752,093	66,500,000
J1060 POI - West Memphis 500 kV	120.0	9.8%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281	17,219	1,000,000

			J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465		
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9.3.33. J1361

This generator is determined to be deliverable up to 40.4 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 38 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 38: NRIS Results for J1361

J1361 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			40.4 MW (80.72%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
J1060 POI - West Memphis 500 kV	50.0	6.29%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	4,799	1,000,000

9.3.34. J1364

This generator is determined to be deliverable up to 19.1 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 39 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 39: NRIS Results for J1364

J1364 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			19.1 MW (12.7%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
West Memphis - Birmingham Steel 500 kV	105.9	39.9%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	3,403,020	31,500,000
Sans Souci - Driver 500 kV	123.8	38.2%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278	378,053	3,300,000

			J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469		
El Dorado - Mt. Olive 500 kV	150.0	6.7%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1434 J1236 J1437 J1469	99,797	2,000,000

9.3.35. J1367

This generator is determined to be deliverable up to 12.4 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 40 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 40: NRIS Results for J1367

J1367 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			12.4 MW (13.1%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Big Cajun - Fancy Point 500 kV	38.8	22.6%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	794,341	29,600,000
Champagne - Bobcat 138 kV	43.9	5.8%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	298,960	198,750,000
J683_Tap - Wilbert 138 kV	45.2	5.7%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	297,920	198,750,000
Webre - Bayou Labutte 500 kV	45.9	21.6%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	2,274,961	89,900,000
Bobcat - Krotz Spring 138 kV	56.4	5.8%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	298,960	198,750,000
Livonia - J683_Tap 138 kV	59.1	5.8%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	298,960	198,750,000
Bayou Labutte - Willow Glen 500 kV	61.6	19.2%	J1205 J1217 J1220 J1236 J1238 J1246 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462 J1465	435,359	26,700,000
Krotz Spring - Line-642 Tap 138 kV	66.0	5.8%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	298,960	198,750,000
Line-642 Tap - Livonia 138 kv	76.7	5.8%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286	298,960	198,750,000

			J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462		
J1060 POI - West Memphis 500 kV	87.5	8.6%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	12,030	1,000,000
Waterford - Tezcucu 230 kV	93.5	5.8%	J1367 J1431 J1441 J1455	2,791,474	38,500,000
Bayou Vista - Terrebonne 230 kV	95.0	32.0%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	1,665,339	198,750,000

9.3.36. J1368

This generator is determined to be deliverable up to 56.4 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 41 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 41: NRIS Results for J1368

J1368 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)			56.4 MW (28.2%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
J639_Tap - Addis 230 kV	95.6	29.8%	J1368 J1372 J1463	23,955,599	276,850,000
J1368 POI - Fancy Point 230 kV	116.3	100.0%	J1368, J1372	13,019,271	78,100,000
J1368 POI - Waterloo 230 kV	161.5	49.2%	J1372, J1368	4,666,667	7,000,000
J1060 POI - West Memphis 500 kV	190.6	7.3%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	21,899	1,000,000
Big Cajun - J639_Tap 230 kV	200.0	100.0%	J1368 J1372	13,019,271	78,100,000

9.3.37. J1372

This generator is determined to be deliverable up to 28.2 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 42 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 42: NRIS Results for J1372

J1372 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)	28.2 MW (28.2%)
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Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
J639_Tap - Addis 230 kV	47.8	29.8%	J1368 J1372 J1463	11,977,799	276,850,000
J1368 POI - Fancy Point 230 kV	58.1	100.0%	J1368 J1372 J1463	6,509,635	78,100,000
J1368 POI - Waterloo 230 kV	80.7	49.2%	J1372, J1368	2,333,333	7,000,000
J1060 POI - West Memphis 500 kV	95.3	7.3%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	10,950	1,000,000
Big Cajun - J639_Tap 230 kV	100.0	100.0%	J1368 J1372	6,509,635	78,100,000

9.3.38. J1373

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 43 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 43: NRIS Results for J1373

J1373 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Ray Braswell - Franklin 500 kV	12.1	5.0%	J1215 J1239 J1245 J1260 J1261 J1280 J1281 J1287 J1293 J1294 J1347 J1351 J1373 J1404 J1425 J1434 J1436 J1439 J1440 J1458	68,379	3,000,000
West Memphis - Birmingham Steel 500 kV	53.8	19.6%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	1,057,637	31,500,000
Perryville - Baxter Wilson 500 kV	67.1	6.3%	J1215 J1239 J1245 J1258 J1260 J1261 J1280 J1281 J1373 J1415 J1425 J1436 J1437 J1452	37,340	1,000,000
Sans Souci - Driver 500 kV	76.7	21.3%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	133,444	3,300,000
J1060 POI - West Memphis 500 kV	78.4	13.2%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281	17,396	1,000,000

			J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465		
El Dorado - Mt. Olive 500 kV	82.0	9.0%	J1261, J1280, J1293, J1373, J1434	85,540	2,000,000
Wynne South - Wynne East 161 kV	95.0	5.2%	J1261 J1280 J1293 J1373 J1434	656,501	6,100,000

9.3.39. J1380

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 44 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 44: NRIS Results for J1380

J1380 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Nelson - Goosport 138 kV #2	0.0	5.1%	J1277 J1279 J1282 J1380 J1424	554,142	162,000,000
Nelson - LC Bulk South 138 kV	0.0	5.1%	J1277 J1279 J1282 J1380 J1424	555,238	162,000,000
Goosport - LC Bulk North 138 kV	0.0	5.1%	J1277 J1279 J1282 J1380 J1424	554,142	162,000,000
LC Bulk North - Henning 138 kV	0.0	6.5%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	708,558	162,000,000
LC Bulk South - Lowegrout 138 kV	6.2	5.9%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	648,325	162,000,000
Cocodrie - Ville Platte 230 kV	12.5	12.0%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	1,849,589	22,600,000
J1384 POI - J1217 POI 230 kV	25.5	12.0%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1465	225,960	4,050,000
West Memphis - Birmingham Steel 500 kV	26.2	5.5%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	629,999	31,500,000
Big Cajun - Fancy Point 500 kV	38.3	30.8%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	2,275,094	29,600,000
Ville Platte - J1384 POI 230 kV	42.0	12.0%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	2,616,733	360,750,000
Lowegrout - Jennings 138 kV	53.7	5.9%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	648,325	162,000,000
Manena - Sadie 230 kV	81.7	19.9%	J1205 J1282 J1380 J1424	1,249,280	9,600,000
Champagne - Bobcat 138 kV	81.7	7.3%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	803,430	198,750,000

Bayou Cove - Richard 138 kV #1	92.3	5.8%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	638,469	162,000,000
J683_Tap - Wilbert 138 kV	95.1	7.3%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	802,336	198,750,000
Webre - Bayou Labutte 500 kV	96.6	31.4%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	6,948,054	89,900,000
Bobcat - Krotz Spring 138 kV	108.2	7.3%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	803,430	198,750,000
Rork - Sellers Rd 230 kV	111.3	11.4%	J1205 J1277 J1278 J1279 J1282 J1380 J1424 J1442	3,585,450	55,000,000
Henning - Hebert 138 kV	118.6	6.5%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	708,558	162,000,000
Livonia - J683_Tap 138 kV	124.5	7.3%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	803,430	198,750,000
Bayou Labutte - Willow Glen 500 kV	129.7	26.2%	J1205 J1217 J1220 J1236 J1238 J1246 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462 J1465	1,250,574	26,700,000
Krotz Spring - Line-642 Tap 138 kV	130.2	7.3%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	803,430	198,750,000
Cooper 230/138 kV Transformer	135.2	44.7%	J1277 J1279 J1380 J1242 J1424 J1442	4,890,910	162,000,000
Hebert - Bayou Cove 138 kV	139.0	6.5%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	708,558	162,000,000
Line-642 Tap - Livonia 138 kv	155.7	7.3%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	803,430	198,750,000
Nelson 500/230 kV Transformer	161.5	31.5%	J1205 J1277 J1278 J1279 J1282 J1380 J1424 J1442 J1452	3,449,701	162,000,000
J1060 POI - West Memphis 500 kV	168.2	10.9%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	31,644	1,000,000
J1380 POI - Cooper 230 kV	178.6	44.7%	J1277 J1279 J1282 J1380 J1424 J1442	4,890,910	162,000,000
Bayou Cove - Richard 138 kV #2	182.6	6.6%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	718,414	162,000,000
Nelson - Richard 500 kV	182.9	29.5%	J1205 J1277 J1278 J1279 J1380 J1282 J1424 J1442 J1452	3,228,482	162,000,000
Richard - Wells 500 kV	189.7	31.3%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442	3,428,894	162,000,000

			J1452 J1462		
Sadie - Rose Bluff 230 kV	196.8	19.9%	J1205 J1282 J1380 J1424	1,249,280	9,600,000
Bayou Vista - Terrebonne 230 kV	200.0	13%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	1,383,564	198,750,000

9.3.40. J1384

This generator is determined to be deliverable up to 12.5 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 45 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 45: NRIS Results for J1384

J1384 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			12.5 MW (6.2%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
J1384 POI - J1217 POI 230 kV	26.2	58.0%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1465	1,289,011	4,050,000
Big Cajun - Fancy Point 500 kV	81.7	35.6%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	2,633,813	29,600,000
Champagne - Bobcat 138 kV	92.3	9.9%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	1,080,362	198,750,000
J683_Tap - Wilbert 138 kV	95.1	9.9%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	1,078,173	198,750,000
Webre - Bayou Labutte 500 kV	96.6	36.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	8,014,087	89,900,000
Bobcat - Krotz Spring 138 kV	104.8	9.9%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	1,080,362	198,750,000
Champagne - E Opelusas 138 kV	118.6	12.1%	J1205 J1217 J1220 J1236 J1238 J1384 J1465	1,328,834	198,750,000
Livonia - J683_Tap 138 kV	120.4	9.9%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	1,080,362	198,750,000
East Leesville - Rodemacher 230 kV	121.6	14.3%	J1217 J1220 J1236 J1238 J1384	16,788,763	67,408,000
E Opelusas - Colton 138 kV	124.5	12.1%	J1205 J1217 J1220 J1236 J1238 J1384 J1465	1,328,834	198,750,000
Bayou Labutte - Willow Glen 500 kV	125.2	30.4%	J1205 J1217 J1220 J1236 J1238 J1246 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	1,452,711	26,700,000

			J1465		
Cocodrie - Coughlin 138 kV	129.7	32.8%	J1217 J1220 J1236 J1238 J1384	3,590,260	198,750,000
Krotz Spring - Line-642 Tap 138 kV	139.0	9.9%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	1,080,362	198,750,000
Line-642 Tap - Livonia 138 kv	148.0	9.9%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	1,080,362	198,750,000
Colton - Bloomfield 138 kV	155.6	12.1%	J1205 J1217 J1220 J1236 J1238 J1384 J1465	1,328,834	198,750,000
Plaisance - Champagne 138 kV	161.5	21.7%	J1205 J1217 J1220 J1236 J1238 J1286 J1290 J1384	2,371,980	198,750,000
J1060 POI - West Memphis 500 kV	180.7	9.7%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	28,297	1,000,000
Manuel - Eunice 138 kV	182.9	14.4%	J1205 J1217 J1220 J1236 J1238 J1286 J1290 J1384	1,578,401	198,750,000
Coughlin - Manuel RCT 138 kV	182.9	14.4%	J1205 J1217 J1220 J1236 J1238 J1384	1,578,401	198,750,000
Manuel RCT - Manuel 138 kV	182.9	14.4%	J1205 J1217 J1220 J1236 J1238 J1384	1,578,401	198,750,000
Richard - Wells 500 kV	193.3	17.3%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1462	1,898,978	162,000,000
Bloomfield - Vatican 138 kV	196.8	8.1%	J1217 J1220 J1236 J1238 J1384 J1462 J1465	3,432,847	217,000,000
Bayou Vista - Terrebonne 230 kV	200.0	15.7%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	1,718,509	198,750,000

9.3.41. J1400

This generator is determined to be deliverable up to 19.1 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 46 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 46: NRIS Results for J1400

J1400 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)		19.1 MW (12.7%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
West Memphis - Birmingham Steel 500 kV	105.9	39.9%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261	3,403,020	31,500,000

			J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469		
Sans Souci - Driver 500 kV	123.8	38.2%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	378,053	3,300,000
El Dorado - Mt. Olive 500 kV	150.0	6.7%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1434 J1236 J1437 J1469	99,797	2,000,000

9.3.42. J1404

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 47 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 47: NRIS Results for J1404

J1404 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Ray Braswell - Franklin 500 kV	0.0	11.9%	J1215 J1239 J1245 J1260 J1261 J1280 J1281 J1287 J1293 J1294 J1347 J1351 J1373 J1404 J1425 J1434 J1436 J1439 J1440 J1458	286,323	3,000,000
Bogalusa - Adams Creek 230 kV	55.8	6.8%	J1239 J1260 J1281 J1294 J1347 J1351 J1404 J1436 J1458	4,129,508	66,500,000
Bogalusa 500/230 kV Transformer	78.9	6.8%	J1239 J1260 J1281 J1294 J1347 J1351 J1404 J1436 J1458	4,129,508	66,500,000
Delta SES - Cleveland 115 kV	138.2	29.2%	J1404 J1439 J1440	4,925,394	6,300,000
Horn Lake - Freeport 161 kV	138.2	5.1%	J1215 J1245 J1261 J1280 J1287 J1293 J1404 J1434 J1439 J1440	1,229,626	10,100,000
J1440 POI - Tunica 115 kV	167.0	14.2%	J1294 J1404 J1439 J1440	3,691,881	28,600,000

9.3.43. J1414

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 48 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 48: NRIS Results for J1414

J1414 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)	0 MW (0%)
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Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
J1007 POI - J1469 POI 161 kV	19.1	6.3%	J1250 J1267 J1414	484,998	1,900,000
West Memphis - Birmingham Steel 500 kV	105.9	18.5%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	1,577,842	31,500,000
Sans Souci - Driver 500 kV	123.8	22.6%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	223,269	3,300,000
El Dorado - Mt. Olive 500 kV	150.0	6.4%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1434 J1236 J1437 J1469	95,445	2,000,000

9.3.44. J1415

This generator is determined to be deliverable up to 31.8 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 49 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 49: NRIS Results for J1415

J1415 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)		31.8 MW (12.7%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
West Memphis - Birmingham Steel 500 kV	141.5	22.4%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	3,179,848	31,500,000
Perryville - Baxter Wilson 500 kV	176.5	10.1%	J1215 J1239 J1245 J1258 J1260 J1261 J1280 J1281 J1373 J1415 J1425 J1436 J1437 J1452	155,135	1,000,000
Sans Souci - Driver 500 kV	201.8	21.4%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	352,652	3,300,000
J1060 POI - West Memphis 500 kV	206.3	28.1%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442	103,572	1,000,000

			J1452 J1455 J1458 J1462 J1463 J1465		
El Dorado - Mt. Olive 500 kV	250.0	12.4%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1434 J1236 J1437 J1469	310,145	2,000,000

9.3.45. J1421

This generator is determined to be deliverable up to 242.2 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 50 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 50: NRIS Results for J1421

J1421 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			242.2 MW (80.73%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
J1060 POI - West Memphis 500 kV	300.0	7.1%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	32,246	1,000,000

9.3.46. J1424

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 51 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 51: NRIS Results for J1424

J1424 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Nelson - Goosport 138 kV #2	0.0	5.7%	J1277 J1279 J1282 J1380 J1424	623,137	162,000,000
Nelson - LC Bulk South 138 kV	0.0	5.7%	J1277 J1279 J1282 J1380 J1424	625,327	162,000,000
Goosport - LC Bulk North 138 kV	0.0	5.7%	J1277 J1279 J1282 J1380 J1424	623,137	162,000,000
LC Bulk North - Henning 138 kV	0.0	7.6%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	830,119	162,000,000
LC Bulk South - Lowegrout 138 kV	6.2	6.9%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	760,029	162,000,000
Cocodrie - Ville Platte 230 kV	12.5	9.6%	J1205 J1220 J1238 J1277 J1278	1,488,958	22,600,000

			J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465		
J1384 POI - J1217 POI 230 kV	25.5	9.6%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1465	181,902	4,050,000
West Memphis - Birmingham Steel 500 kV	26.2	5.4%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	616,353	31,500,000
Big Cajun - Fancy Point 500 kV	38.3	31.9%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	2,357,932	29,600,000
Ville Platte - J1384 POI 230 kV	42.0	9.6%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	2,106,525	360,750,000
Lowegrout - Jennings 138 kV	53.7	6.9%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	760,029	162,000,000
Manena - Sadie 230 kV	81.7	24.7%	J1205 J1282 J1380 J1424	1,545,623	9,600,000
Champagne - Bobcat 138 kV	81.7	7.3%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	793,579	198,750,000
Bayou Cove - Richard 138 kV #1	92.3	6.8%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	747,983	162,000,000
J683_Tap - Wilbert 138 kV	95.1	7.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	791,390	198,750,000
Webre - Bayou Labutte 500 kV	96.6	32.5%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	7,202,926	89,900,000
Bobcat - Krotz Spring 138 kV	108.2	7.3%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	793,579	198,750,000
Rork - Sellers Rd 230 kV	111.3	13.4%	J1205 J1277 J1278 J1279 J1282 J1380 J1424 J1442	4,238,210	55,000,000
Henning - Hebert 138 kV	118.6	7.6%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	830,119	162,000,000
Livonia - J683_Tap 138 kV	119.9	7.3%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	793,579	198,750,000
Scott - Bonin 138 kV	124.5	5.4%	J1282 J1286 J1290 J1424 J1462 J1465	595,758	162,000,000
Bayou Labutte - Willow Glen 500 kV	129.7	27.1%	J1205 J1217 J1220 J1236 J1238 J1246 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462 J1465	1,295,493	26,700,000
Krotz Spring - Line-642 Tap 138 kV	130.2	7.3%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	793,579	198,750,000
Cooper 230/138 kV Transformer	135.2	26.4%	J1277 J1279 J1380 J1242 J1424	2,885,703	162,000,000

			J1442		
Hebert - Bayou Cove 138 kV	139.0	7.6%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	830,119	162,000,000
Line-642 Tap - Livonia 138 kv	155.7	7.3%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	793,579	198,750,000
Nelson 500/230 kV Transformer	161.5	37.7%	J1205 J1277 J1278 J1279 J1282 J1380 J1424 J1442 J1452	4,123,214	162,000,000
J1060 POI - West Memphis 500 kV	168.2	10.8%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	31,430	1,000,000
J1380 POI - Cooper 230 kV	178.6	26.4%	J1277 J1279 J1282 J1380 J1424 J1442	2,885,703	162,000,000
Bayou Cove - Richard 138 kV #2	182.6	7.7%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	842,165	162,000,000
Nelson - Richard 500 kV	182.9	35.1%	J1205 J1277 J1278 J1279 J1380 J1282 J1424 J1442 J1452	3,842,858	162,000,000
Richard - Wells 500 kV	189.7	34.0%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1462	3,807,813	162,000,000
Sadie - Rose Bluff 230 kV	196.8	24.7%	J1205 J1282 J1380 J1424	1,545,623	9,600,000
Bayou Vista - Terrebonne 230 kV	200.0	13.0%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	1,418,590	198,750,000

9.3.47. J1425

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 52 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 52: NRIS Results for J1425

J1421 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Ray Braswell - Franklin 500 kV	14.5	5.6%	J1215 J1239 J1245 J1260 J1261 J1280 J1281 J1287 J1293 J1294 J1347 J1351 J1373 J1404 J1425 J1434 J1436 J1439 J1440 J1458	91,737	3,000,000
West Memphis - Birmingham Steel 500 kV	64.5	14.1%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373	916,547	31,500,000

			J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469		
Perryville - Baxter Wilson 500 kV	80.5	10.0%	J1215 J1239 J1245 J1258 J1260 J1261 J1280 J1281 J1373 J1415 J1425 J1436 J1437 J1452	69,995	1,000,000
Sans Souci - Driver 500 kV	85.8	13.8%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	103,872	3,300,000
J620_Tap - Macon Lake 115 kV	92.0	5.6%	J1260 J1281 J1425	787,546	7,900,000
J1060 POI - West Memphis 500 kV	94.1	19.1%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	30,260	1,000,000
El Dorado - Mt. Olive 500 kV	114.0	10.9%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1434 J1236 J1437 J1469	124,090	2,000,000

9.3.48. J1431

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 53 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 53: NRIS Results for J1431

J1431 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Tezcuco - Frisco 230 kV #1	0.0	12.2%	J1431 J1441	1,007,028	2,000,000
Tezcuco - Frisco 230 kV #2	0.0	12.1%	J1431 J1441	1,006,645	2,000,000
Gypsy - Ironman 230 kV	46.3	5.0%	J1431 J1441	249,008	500,000
Frisco - Convent 230 kV	161.5	8.3%	J1431 J1441	23,153,240	46,000,000
J1060 POI - West Memphis 500 kV	184.3	7.3%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	21,872	1,000,000
Waterford - Tezcuco 230 kV	200.0	16.3%	J1367 J1431 J1441 J1455	16,382,551	38,500,000

9.3.49. J1434

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 54 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 54: NRIS Results for J1434

J1434 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Ray Braswell - Franklin 500 kV	81.3	6.2%	J1215 J1239 J1245 J1260 J1261 J1280 J1281 J1287 J1293 J1294 J1347 J1351 J1373 J1404 J1425 J1434 J1436 J1439 J1440 J1458	89,252	3,000,000
Wabbaseka - Humphrey 115 kV	82.5	7.1%	J1215 J1245 J1261 J1280 J1293 J1434	885,990	10,300,000
El Dorado - Mt. Olive 500 kV	82.8	6.7%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1434 J1236 J1437 J1469	66,531	2,000,000
Horn Lake - Freeport 161 kV	83.9	6.4%	J1215 J1245 J1261 J1280 J1287 J1293 J1404 J1434 J1439 J1440	915,365	10,100,000
Wabbaseka - Wabbaseka West 115 kV	86.3	7.1%	J1215 J1245 J1261 J1280 J1293 J1434	343,858	4,000,000
Wynne South - Wynne East 161 kV	93.5	12.9%	J1261, J1280, J1293, J1373, J1434	1,735,044	6,100,000
Crossroads - Clarkmun 230/115 kV	96.0	8.1%	J1215 J1245 J1261 J1280 J1293 J1434	1,345,217	6,000,000
Wabbaseka West - Alzheimer 115 kV	98.5	7.1%	J1215 J1245 J1261 J1280 J1293 J1434	584,559	6,800,000
Alzheimer - Woodward 115 kV	100.0	7.1%	J1215 J1245 J1261 J1280 J1293 J1434	1,891,220	22,000,000

9.3.50. J1436

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 55 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 55: NRIS Results for J1436

J1436 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Ray Braswell - Franklin 500 kV	0.0	9.9%	J1215 J1239 J1245 J1260 J1261 J1280 J1281 J1287 J1293 J1294 J1347 J1351 J1373 J1404 J1425 J1434 J1436 J1439 J1440 J1458	71,330	3,000,000

Bogalusa - Adams Creek 230 kV	6.4	8.7%	J1239 J1260 J1281 J1294 J1347 J1351 J1404 J1436 J1458	1,597,219	66,500,000
West Memphis - Birmingham Steel 500 kV	16.7	6.2%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	175,126	31,500,000
Bogalusa 500/230 kV Transformer	28.3	8.7%	J1239 J1260 J1281 J1294 J1347 J1351 J1404 J1436 J1458	1,597,219	66,500,000
Perryville - Baxter Wilson 500 kV	35.3	21.5%	J1215 J1239 J1245 J1258 J1260 J1261 J1280 J1281 J1373 J1415 J1425 J1436 J1437 J1452	64,972	1,000,000
Sans Souci - Driver 500 kV	40.4	5.6%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	18,309	3,300,000
J1060 POI - West Memphis 500 kV	48.5	12.0%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	8,748	1,000,000
El Dorado - Mt. Olive 500 kV	50.0	8.7%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1434 J1236 J1437 J1469	43,270	2,000,000

9.3.51. J1437

This generator is determined to be deliverable up to 22.9 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 56 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 56: NRIS Results for J1437

J1437 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)		22.9 MW (12.7%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
West Memphis - Birmingham Steel 500 kV	101.9	11.5%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	1,178,008	31,500,000
Perryville - Baxter Wilson 500 kV	127.1	5.5%	J1215 J1239 J1245 J1258 J1260	62,495	1,000,000

			J1261 J1280 J1281 J1373 J1415 J1425 J1436 J1437 J1452		
Sans Souci - Driver 500 kV	145.3	12.5%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	148,094	3,300,000
J1060 POI - West Memphis 500 kV	148.5	6.5%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	16,360	1,000,000
El Dorado - Mt. Olive 500 kV	180.0	6.1%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1434 J1236 J1437 J1469	110,572	2,000,000

9.3.52. J1439

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 57 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 57: NRIS Results for J1439

J1439 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Ray Braswell - Franklin 500 kV	18.9	7.4%	J1215 J1239 J1245 J1260 J1261 J1280 J1281 J1287 J1293 J1294 J1347 J1351 J1373 J1404 J1425 J1434 J1436 J1439 J1440 J1458	42,438	3,000,000
Delta SES - Cleveland 115 kV	33.1	6.8%	J1404 J1439 J1440	274,921	6,300,000
J1440 POI - Tunica 115 kV	33.1	76.3%	J1294 J1404 J1439 J1440	4,749,364	28,600,000
Horn Lake - Freeport 161 kV	40.0	12.4%	J1215 J1245 J1261 J1280 J1287 J1293 J1404 J1434 J1439 J1440	708,226	10,100,000

9.3.53. J1440

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 58 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 58: NRIS Results for J1440

J1440 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)	0 MW (0%)
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Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Ray Braswell - Franklin 500 kV	75.6	7.4%	J1215 J1239 J1245 J1260 J1261 J1280 J1281 J1287 J1293 J1294 J1347 J1351 J1373 J1404 J1425 J1434 J1436 J1439 J1440 J1458	169,752	3,000,000
Delta SES - Cleveland 115 kV	132.4	6.8%	J1404 J1439 J1440	1,099,685	6,300,000
Horn Lake - Freeport 161 kV	132.5	12.4%	J1215 J1245 J1261 J1280 J1287 J1293 J1404 J1434 J1439 J1440	2,832,904	10,100,000
J1440 POI - Tunica 115 kV	160.0	76.3%	J1294 J1404 J1439 J1440	18,997,457	28,600,000

9.3.54. J1441

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 59 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 59: NRIS Results for J1441

J1441 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)			0 MW (0%)		
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Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Tezcuco - Frisco 230 kV #1	0.0	12.0%	J1431 J1441	992,972	2,000,000
Tezcuco - Frisco 230 kV #2	0.0	12.0%	J1431 J1441	993,355	2,000,000
Gypsy - Ironman 230 kV	46.3	5.1%	J1431 J1441	250,992	500,000
Frisco - Convent 230 kV	161.5	8.2%	J1431 J1441	22,846,760	46,000,000
J1060 POI - West Memphis 500 kV	184.3	7.5%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	22,542	1,000,000
Waterford - Tezcuco 230 kV	200.0	15.1%	J1367 J1431 J1441 J1455	15,144,803	38,500,000

9.3.55. J1442

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 60 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 60: NRIS Results for J1442

J1442 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)			0 MW (0%)		
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Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
LC Bulk North - Henning 138 kV	0.0	6.4%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	577,332	162,000,000
LC Bulk South - Lowegrout 138 kV	5.1	5.9%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	528,544	162,000,000
Cocodrie - Ville Platte 230 kV	10.3	7.4%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	942,362	22,600,000
J1384 POI - J1217 POI 230 kV	21.0	7.4%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1465	115,126	4,050,000
West Memphis - Birmingham Steel 500 kV	21.6	6.5%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	605,124	31,500,000
Big Cajun - Fancy Point 500 kV	31.6	27.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	1,659,114	29,600,000
Ville Platte - J1384 POI 230 kV	34.7	7.4%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	1,333,220	360,750,000
Lowegrout - Jennings 138 kV	67.4	5.9%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	528,544	162,000,000
Champagne - Bobcat 138 kV	67.4	6.0%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	541,823	198,750,000
Bayou Cove - Richard 138 kV #1	76.2	5.8%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	520,412	162,000,000
J683_Tap - Wilbert 138 kV	78.5	6.0%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	540,920	198,750,000
Webre - Bayou Labutte 500 kV	79.7	27.8%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	5,077,564	89,900,000
Bobcat - Krotz Spring 138 kV	88.2	6.0%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	541,823	198,750,000
Sabine - Mud Lake 230 kV	89.3	16.8%	J1277 J1278 J1279 J1442	133,812	400,000
Rork - Sellers Rd 230 kV	91.8	12.3%	J1205 J1277 J1278 J1279 J1282 J1380 J1424 J1442	3,207,748	55,000,000
Henning - Hebert 138 kV	97.9	6.4%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	577,332	162,000,000
Livonia - J683_Tap 138 kV	102.0	6.0%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	541,823	198,750,000
J1282 POI - J1424 POI 230 kV	102.7	8.7%	J1277, J1278, J1279, J1282, J1442	390,000	4,800,000
Bayou Labutte - Willow Glen 500 kV	107.0	23.1%	J1205 J1217 J1220 J1236 J1238 J1246 J1277 J1278 J1279 J1282	910,298	26,700,000

			J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462 J1465		
Krotz Spring - Line-642 Tap 138 kV	107.4	6.0%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	541,823	198,750,000
Cooper 230/138 kV Transformer	111.6	6.5%	J1277 J1279 J1380 J1242 J1424 J1442	589,981	162,000,000
Hebert - Bayou Cove 138 kV	114.7	6.4%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	577,332	162,000,000
Line-642 Tap - Livonia 138 kV	115.8	6.0%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	541,823	198,750,000
Pelican Road - Jacinto 138 kV	116.5	72.5%	J1278 J1442	12,696,465	13,700,000
Sans Souci - Driver 500 kV	119.2	5.9%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	64,012	3,300,000
J1278 POI - Moscow 138 kV	128.5	7.6%	J1278 J1442	1,484,996	7,000,000
Nelson 500/230 kV Transformer	133.2	25.9%	J1205 J1277 J1278 J1279 J1282 J1380 J1424 J1442 J1452	2,339,144	162,000,000
J1060 POI - West Memphis 500 kV	138.7	11.5%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	27,299	1,000,000
J1380 POI - Cooper 230 kV	147.4	6.5%	J1277 J1279 J1282 J1380 J1424 J1442	589,981	162,000,000
Bayou Cove - Richard 138 kV #2	150.6	6.5%	J1277 J1278 J1279 J1282 J1380 J1424 J1442	586,367	162,000,000
Nelson - Richard 500 kV	150.9	33.0%	J1205 J1277 J1278 J1279 J1380 J1282 J1424 J1442 J1452	2,977,914	162,000,000
Richard - Wells 500 kV	154.1	29.9%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1462	2,835,162	162,000,000
Moscow - Corrigan 138 kV	162.3	7.6%	J1278 J1442	2,163,851	10,200,000
Bayou Vista - Terrebonne 230 kV	165.0	11.0%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	994,245	198,750,000

9.3.56. J1452

This generator is determined to be deliverable up to 6.2 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 61 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 61: NRIS Results for J1452

J1452 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			6.2 MW (3.1%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Cocodrie - Ville Platte 230 kV	12.4	9.1%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	1,405,378	22,600,000
J1384 POI - J1217 POI 230 kV	25.5	9.1%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1465	171,692	4,050,000
West Memphis - Birmingham Steel 500 kV	26.2	8.8%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	999,584	31,500,000
Big Cajun - Fancy Point 500 kV	38.3	15.1%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	1,114,619	29,600,000
Ville Platte - J1384 POI 230 kV	95.1	9.1%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	1,988,279	360,750,000
Webre - Bayou Labutte 500 kV	113.2	15.5%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	3,430,809	89,900,000
Perryville - Baxter Wilson 500 kV	124.5	9.3%	J1215 J1239 J1245 J1258 J1260 J1261 J1280 J1281 J1373 J1415 J1425 J1436 J1437 J1452	114,461	1,000,000
Bayou Labutte - Willow Glen 500 kV	141.2	12.8%	J1205 J1217 J1220 J1236 J1238 J1246 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462 J1465	610,712	26,700,000
Sans Souci - Driver 500 kV	155.7	8.3%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	108,996	3,300,000
J1220 - Rapides 230 kV	147.2	6.3%	J1220, J1238, J1452	1,347,053	10,000,000
Nelson 500/230 kV Transformer	161.5	9.2%	J1205 J1277 J1278 J1279 J1282 J1380 J1424 J1442 J1452	1,010,817	162,000,000
J1060 POI - West Memphis 500 kV	181.9	13.3%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	38,149	1,000,000
Rapides - Forest Hill 230 kV	182.5	6.1%	J1452, J1220, J1238	5,166,384	42,768,000

Nelson - Richard 500 kV	182.9	11.3%	J1205 J1277 J1278 J1279 J1380 J1282 J1424 J1442 J1452	1,235,322	162,000,000
Richard - Wells 500 kV	196.8	14.1%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1462	1,545,247	162,000,000
Bayou Vista - Terrebonne 230 kV	200.0	6.43%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	703,822	198,750,000

9.3.57. J1455

This generator is determined to be deliverable up to 19.6 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 55 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 62: NRIS Results for J1455

J1455 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			19.6 MW (13.1%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Big Cajun - Fancy Point 500 kV	61.2	23.5%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	1,300,820	29,600,000
Champagne - Bobcat 138 kV	69.2	6.0%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	493,387	198,750,000
J683_Tap - Wilbert 138 kV	71.3	6.0%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	491,745	198,750,000
Webre - Bayou Labutte 500 kV	72.4	22.5%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	3,744,968	89,900,000
Bobcat - Krotz Spring 138 kV	89.0	6.0%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	493,387	198,750,000
Livonia - J683_Tap 138 kV	93.4	6.0%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	493,387	198,750,000
Bayou Labutte - Willow Glen 500 kV	97.2	19.9%	J1205 J1217 J1220 J1236 J1238 J1246 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462 J1465	713,214	26,700,000
Krotz Spring - Line-642 Tap 138 kV	104.3	6.0%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424	493,387	198,750,000

			J1442 J1452 J1455 J1462		
Line-642 Tap - Livonia 138 kv	121.1	6.0%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	493,387	198,750,000
J1060 POI - West Memphis 500 kv	138.2	8.6%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	19,075	1,000,000
Waterford - Tezcucu 230 kv	147.6	5.5%	J1367 J1431 J1441 J1455	4,181,173	38,500,000
Bayou Vista - Terrebonne 230 kv	150.0	30.9%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	2,536,716	198,750,000

9.3.58. J1458

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 63 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 63: NRIS Results for J1458

J1458 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)			0 MW (0%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Ray Braswell - Franklin 500 kv	0.0	16.3%	J1215 J1239 J1245 J1260 J1261 J1280 J1281 J1287 J1293 J1294 J1347 J1351 J1373 J1404 J1425 J1434 J1436 J1439 J1440 J1458	469,582	3,000,000
Bogalusa - Adams Creek 230 kv	66.8	9.3%	J1239 J1260 J1281 J1294 J1347 J1351 J1404 J1436 J1458	6,828,478	66,500,000
Bogalusa 500/230 kv Transformer	113.6	9.3%	J1239 J1260 J1281 J1294 J1347 J1351 J1404 J1436 J1458	6,828,478	66,500,000
Andrus - Refuge 115 kv	116.7	12.4%	J1260 J1458	0	0
Greenville - SE Greenville 115 kv	132.4	6.8%	J1458	5,300,000	5,300,000
Andrus 230/115 kv Transformer	152.6	24.6%	J1260 J1281 J1458	11,683,096	18,000,000
Isola - Belzoni 115 kv	155.8	5.3%	J1294 J1458	269,431	500,000
Refuge - Greenville Industrial 115 kv	192.9	12.4%	J1260 J1458	0	0
J1060 POI - West Memphis 500 kv	200.0	5.1%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463	13,653	1,000,000

			J1465		
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9.3.59. J1462

This generator is determined to be deliverable up to 19.6 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 64 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 64: NRIS Results for J1462

J1462 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)			19.6 MW (13.1%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Big Cajun - Fancy Point 500 kV	61.2	34.8%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	1,930,428	29,600,000
Champagne - Bobcat 138 kV	69.2	11.5%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	943,264	198,750,000
J683_Tap - Wilbert 138 kV	71.3	11.5%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	942,443	198,750,000
Webre - Bayou Labutte 500 kV	72.4	35.1%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	5,841,019	89,900,000
Bobcat - Krotz Spring 138 kV	74.1	11.5%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	943,264	198,750,000
Scanlan - Bosco RCT 138 kV	89.0	66.0%	J1286 J1290 J1462	8,867,075	9,900,000
Livonia - J683_Tap 138 kV	89.9	11.5%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	943,264	198,750,000
Scott - Bonin 138 kV	93.4	29.4%	J1282 J1286 J1290 J1424 J1462 J1465	2,416,434	162,000,000
Bayou Labutte - Willow Glen 500 kV	97.2	29.8%	J1205 J1217 J1220 J1236 J1238 J1246 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462 J1465	1,068,388	26,700,000
Krotz Spring - Line-642 Tap 138 kV	104.3	11.5%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	943,264	198,750,000
Line-642 Tap - Livonia 138 kv	121.1	11.5%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462	943,264	198,750,000
J1060 POI - West Memphis 500 kV	122.6	9.2%	J1205 J1215 J1217 J1219 J1220	20,279	1,000,000

			J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465		
J1462 POI - Scanlan 138 kV	137.2	66.0%	J1286 J1290 J1462	15,609,479	55,000,000
Richard - Wells 500 kV	141.5	14.7%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1462	1,205,753	162,000,000
Bloomfield - Vatican 138 kV	147.6	13.6%	J1217 J1220 J1236 J1238 J1384 J1462 J1465	4,327,172	217,000,000
Bayou Vista - Terrebonne 230 kV	150.0	17.9%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462	1,467,847	198,750,000

9.3.60. J1463

This generator is determined to be deliverable up to 42.3 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 65 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 65: NRIS Results for J1463

J1463 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)			42.3 MW (28.2%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
J639_Tap - Addis 230 kV	71.7	53.3%	J1368 J1372 J1463	17,966,699	276,850,000
J1368 POI - Fancy Point 230 kV	86.1	100.0%	J1368 J1372 J1463	9,760,547	78,100,000
Big Cajun - Waterloo 230 kV	121.1	100.0%	J1463	3,100,000	3,100,000
J1060 POI - West Memphis 500 kV	150.0	7.3%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	16,424	1,000,000

9.3.61. J1465

This generator is determined to be deliverable up to 60.9 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 66 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 66: NRIS Results for J1465

J1465 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)		60.9 MW (40.6%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
J1384 POI - J1217 POI 230 kV	69.2	5.7%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1384 J1424 J1442 J1452 J1465	80,835	4,050,000
J683_Tap - Wilbert 138 kV	72.8	57.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	4,694,976	198,750,000
Cocodrie - Ville Platte 230 kV	78.6	5.7%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	661,673	22,600,000
Champagne - E Opelusas 138 kV	89.0	23.2%	J1205 J1217 J1220 J1236 J1238 J1384 J1465	1,905,410	198,750,000
Livonia - J683_Tap 138 kV	89.9	57.2%	J1205 J1217 J1220 J1236 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1455 J1462 J1465	4,696,618	198,750,000
Scott - Bonin 138 kV	91.2	8.1%	J1282 J1286 J1290 J1424 J1462 J1465	666,942	162,000,000
E Opelusas - Colton 138 kV	93.4	23.2%	J1205 J1217 J1220 J1236 J1238 J1384 J1465	1,905,410	198,750,000
Bayou Labutte - Willow Glen 500 kV	111.0	5.5%	J1205 J1217 J1220 J1236 J1238 J1246 J1277 J1278 J1279 J1282 J1286 J1290 J1367 J1380 J1384 J1424 J1442 J1452 J1455 J1462 J1465	198,195	26,700,000
Colton - Bloomfield 138 kV	121.1	23.2%	J1205 J1217 J1220 J1236 J1238 J1384 J1465	1,905,410	198,750,000
J1060 POI - West Memphis 500 kV	121.9	8.3%	J1205 J1215 J1217 J1219 J1220 J1223 J1236 J1238 J1239 J1245 J1246 J1257 J1258 J1259 J1260 J1277 J1278 J1279 J1280 J1281 J1282 J1286 J1290 J1351 J1361 J1367 J1368 J1372 J1373 J1380 J1384 J1415 J1421 J1424 J1425 J1431 J1436 J1437 J1441 J1442 J1452 J1455 J1458 J1462 J1463 J1465	18,392	1,000,000
Ville Platte - J1384 POI 230 kV	141.5	5.7%	J1205 J1220 J1238 J1277 J1278 J1279 J1282 J1286 J1290 J1380 J1424 J1442 J1452 J1465	936,112	360,750,000
Bloomfield - Vatican 138 kV	150.0	14.5%	J1217 J1220 J1236 J1238 J1384 J1462 J1465	4,633,069	217,000,000

9.3.62. J1469

This generator is determined to be deliverable up to 25.5 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 67 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 67: NRIS Results for J1469

J1469 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)		25.5 MW (12.7%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	DFAX	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
West Memphis - Birmingham Steel 500 kV	141.2	39.9%	J1205 J1215 J1220 J1238 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1282 J1293 J1364 J1373 J1380 J1400 J1414 J1415 J1424 J1425 J1436 J1437 J1442 J1452 J1469	4,539,635	31,500,000
Sans Souci - Driver 500 kV	165.0	38.3%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1277 J1278 J1279 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1436 J1437 J1442 J1452 J1469	505,655	3,300,000
El Dorado - Mt. Olive 500 kV	200.0	6.6%	J1215 J1239 J1245 J1250 J1258 J1260 J1261 J1267 J1280 J1281 J1293 J1364 J1373 J1400 J1414 J1415 J1425 J1434 J1236 J1437 J1469	132,862	2,000,000

10. Shared Network Upgrades Analysis

Shared Network Upgrade (SNU) Analysis, which tests for Network Upgrades driven by higher queued interconnection projects, was performed for this System Impact Study.

The maximum MW impacts and SNU cost allocations appear in Table 68.

Table 68: Maximum MW Impact and SNU Cost Allocations

Network Upgrades	Project Study Cycle	Projects sharing cost	MW Contribution	Total NU Cost (\$)	Cost Responsibility (\$)
Stuttgart Ricuskey – Stuttgart Industrial 115 kV Reconductor	DPP-2014-AUG	J348	13.70	11,400,000	1,602,997
	DPP-2019-1	J1215	16.82		1,968,059
	DPP-2019-1	J1261	24.71		2,891,245
	DPP-2019-1	J1280	42.20		4,937,699
Kuhn Rd - Ebony 161 kV Reconductor	DPP-2017-AUG	J934	133.02	4,900,000	2,688,603
	DPP-2018-APR	J1007	14.67		296,510
	DPP-2018-APR	J1125	23.18		468,515
	DPP-2019-1	J1267	26.98		545,320
	DPP-2019-1	J1364	14.05		283,979
	DPP-2019-1	J1469	14.08		284,585
	DPP-2019-1	J1400	16.45		332,488
Bogalusa – Franklin 500 kV Upgrade	DPP-2017-AUG	J830	5	500,000	25,719
	DPP-2017-AUG	J868	1.756		9,032
	DPP-2017-AUG	J908	3.50		18,003
	DPP-2019-1	J1249	86.95		447,247

11. Cost Allocation



The cost allocation of Network Upgrades (NU) for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of NRIS as of the draft System Impact Study report date.

11.1. Cost Assumptions for Network Upgrades

The cost estimate for each NU identified in the System Impact Study is provided by the corresponding Transmission Owner.

11.2. Cost Allocation Methodology

The costs of NU for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Study Case.

Cost Allocation Methodology for Thermal Constraints

1. With all study group generation projects dispatched in the Study Case, all thermal constraints are identified.
2. Distribution factor from each project on each constraint is obtained.
3. For each thermal constraint, the maximum MW contribution (increasing flow) from each project is then calculated in the Post Case without any Network Upgrades.
4. For each thermal constraint, the cost estimates for one or a subset of NU are provided by the corresponding Transmission Owner.
5. The cost of each NU is allocated based on the pro rata share of the MW contribution from each project on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of one NU is:

$$\text{Project A Cost Portion of NU} = \frac{\text{Max(Proj. A MW contribution on constraint)}}{\sum_i \text{Max(Proj. i MW contrution on constraint)}}$$

6. The total NU costs for each project are calculated if more than one NU is required.



Appendix A – Cost Allocation Summary (CEII)

Appendix B – Network Upgrade Table Per Project (CEII)

Appendix C – MISO ERIS Analysis (CEII)

Appendix D – MISO Deliverability Analysis (CEII)